

Lorenz Energy Cycle Climatology for the Southwestern Atlantic Cyclones

Danilo Couto de Souza^{1,2*}, Pedro Leite da Silva Dias¹,
Carolina Barnez Gramscianinov³, Ricardo Camargo¹

¹Institute of Astronomy, Geophysics and Atmospheric Sciences, São
Paulo University, Rua do Matão, 266, São Paulo, 05508-090, São Paulo,
Brazil.

²Climate Risk Initiative, IRB(re), Av. República do Chile, 330 - 4o
andar, Rio de Janeiro, 20031-170, Rio de Janeiro, Brazil.

³Institute for Coastal Systems Analysis and Modeling,
Helmholtz-Zentrum Hereon, Max-Planck-Straße, 1, Geesthacht, 21502,
Schleswig-Holstein, Germany.

*Corresponding author(s). E-mail(s): danilo.oceano@gmail.com 

Contributing authors: pldsdiass@gmail.com 

cbgramscianinov@gmail.com ; ricamarg@usp.br 

Abstract

This study presents a climatological assessment of the Lorenz Energy Cycle (LEC) applied to South Atlantic cyclones, using a Semi-Lagrangian framework. ~~Over~~ The published analysis included 6,700–789 cyclones ~~were~~ identified from ERA5 reanalysis (~~1979–2020~~1979–2020); the validated corrected rerun currently comprises 3, ~~and~~ 820 of those systems, with LEC components ~~were computed and~~ averaged across four objectively defined life cycle phases: incipient, intensification, mature, and decay. ~~Results reveal a coherent energy transfer structure. The corrected results show that baroclinic conversions dominate during intensification~~ conversion is the dominant conversion pathway, ~~while barotropic conversions peak especially during intensification~~. Barotropic conversion is substantially weaker than reported: it is close to neutral in the ~~mature phase~~ pooled median, changes sign across phases and cases, and does not reach magnitudes ~~2–3~~ 2–3 times larger than ~~its~~ baroclinic ~~counterparts~~ counterpart. Diabatic generation of eddy available potential energy plays a secondary but relevant role, particularly during intensification, ~~occasionally surpassing baroclinic contributions~~.

Eddy kinetic energy imports are most significant during the early phases, reinforcing development. Despite substantial variability among systems, an EOF analysis shows ~~that most cyclones share a common recognizable~~ leading energy structure, ~~with variability manifesting as amplification or suppression of specific pathways~~ whereas higher modes can exchange rank after pattern matching. The leading ~~EOFs are modes~~ remain linked to differences in cyclone intensity, genesis region, and seasonality. Among the most intense systems, ~~distinct five corrected~~ clusters ~~emerge with varying~~ retain differing energetic configurations, ~~some dominated by baroclinic processes, others by barotropic conversions or enhanced diabatic generation~~ although their membership and composition change. These findings demonstrate that South Atlantic cyclones encompass a spectrum of dynamical behaviors and can be classified based on their energy cycle characteristics. This study provides ~~the first a~~ large-sample application of the Semi-Lagrangian LEC to extratropical cyclones in the South Atlantic and highlights the ~~importance~~ dominant baroclinic pathway together with the ~~case-dependent role~~ of barotropic ~~processes in their development~~ conversion. ~~The results offer a robust framework for interpreting cyclone energetics and establish a comprehensive baseline for classifying these systems.~~

Keywords: Cyclone energetics, Baroclinic conversion, Barotropic conversion, Latent heat release, South Atlantic cyclones

Acknowledgements

The first author would like to thank the Laboratory of Applied Meteorology for Regional Meteorological Systems (MASTER) for providing the data processing infrastructure. Special thanks are extended to Jean Peres and Djalma Vieira for their invaluable technical assistance and support in the laboratory, which contributed significantly to the success of this research. CBG acknowledges the Helmholtz European Partnership “Research Capacity Building for Healthy, Productive and Resilient Seas” (SEA-ReCap, grant no. PIE-0025).

1 Introduction

Cyclones play a crucial role in the Earth’s climate system by redistributing heat, moisture, and momentum across different regions. These dynamic weather systems, contribute to the global energy balance by transferring energy from the tropics to higher latitudes. Understanding cyclone behavior is essential for improving climate projections and assessing future climate change impacts. In South America, surface cyclones influence precipitation regimes across the continent (Reboita et al., 2010, 2018; de Souza et al., 2022) often contributing to extreme rainfall events (e.g., de Souza and da Silva, 2021; de Souza et al., 2024), and are associated with severe meteorological hazards such as intense winds (Cardoso et al., 2022; de Souza and da Silva, 2021), high sea waves (da Silva et al., 2025; Gramscianinov et al., 2023), and storm surges along coastal areas (Leal et al., 2023; Tecchio et al., 2024). These extreme

72 events can have profound socioeconomic impacts, including damage to coastal infras-
73 tructure, disruptions to port operations, and interruptions in oil and gas exploration.
74 This is particularly critical in the southeastern region of South America, where large
75 metropolitan areas, such São Paulo, Rio de Janeiro and Buenos Aires are located, as
76 well as the Port of Santos, the largest in Latin America.

77 The Lorenz Energy Cycle (LEC) is a valuable framework for understanding the
78 conversion and flow of energy within the atmosphere, particularly in relation to synop-
79 tic and large-scale processes (Lorenz, 1967). It divides the atmospheric energy budget
80 into four main reservoirs: kinetic and available potential energy, both for zonal (mean
81 flow) and eddy components. Lorenz (1955) formulated the concept of available poten-
82 tial energy (APE), building upon Margules (1903) concept of energy based on the
83 hypothetical adiabatic redistribution of atmospheric mass. The LEC serves as a foun-
84 dational tool for quantifying and understanding atmospheric dynamics, by tracking
85 energy transformations and explaining how these forms of energy are generated by
86 adiabatic processes and dissipated by friction.

87 Substantial efforts have been made to estimate the LEC components for the global
88 circulation (e.g., Muench, 1965; Wiin-Nielsen, 1968; Hu et al., 2004; Li et al., 2007;
89 Oort, 2018), to assess its projected changes under climate change scenarios (e.g.,
90 Hernández-Deckers and von Storch, 2010; Veiga and Ambrizzi, 2013; Pan et al., 2017;
91 Michaelides, 2021), and to explore its relationship with phases of the El Niño–Southern
92 Oscillation (Gutierrez et al., 2009). The LEC framework has also been applied in a
93 range of contexts, from idealized baroclinic wave simulations (Kirshbaum et al., 2018;
94 Rantanen et al., 2019), to rotating annulus experiments (Young, 2014), and even to
95 the atmospheres of other planets (e.g., Read et al., 2020). In contrast, studies exam-
96 ining the LEC in relation to cyclonic systems are more limited and primarily consist
97 of case studies. These have typically focused on tropical (Brennan and Vincent, 1980;
98 Veiga et al., 2008), subtropical (Michaelides, 1987; Dias Pinto et al., 2013; Pezza et al.,
99 2014; Cavicchia et al., 2018), and extratropical cyclones (Michaelides, 1992; Wahab
100 et al., 2002; Bulic, 2006; Pezza et al., 2010; Dias Pinto and Rocha, 2011). Noteworthy
101 contributions include the work of Black and Pezza (2013), which investigated explosive
102 cyclones across all major ocean basins over a 32-year period and identified a univer-
103 sal signature of explosive cyclogenesis, and Okajima et al. (2021), which quantified
104 the contribution of migratory cyclones to global energetics, highlighting their role in
105 accelerating westerly winds worldwide.

106 Although the original formulation by Lorenz (1967) was intended for global-scale
107 studies, analyzing specific atmospheric regions is valuable for understanding the mech-
108 anisms driving the development of individual systems, such as cyclones, and the energy
109 conversions throughout their life cycle. The initial framework for such regional analy-
110 sis was introduced by Muench (1965), who studied Northern Hemisphere stratospheric
111 circulation during winter. Subsequent formulations by Dutton and Johnson (1967),
112 Vincent and Chang (1973), and Smith (1980) refined the approach, but it was only
113 with Brennan and Vincent (1980) that a generalized set of equations, including both
114 eddy and zonal components of kinetic and APE for limited regions in the troposphere,
115 was established. However, cyclones are mobile systems that often traverse large regions
116 (Hoskins and Hodges, 2002; de Souza et al., 2024, e.g.). Since these studies employ the

117 LEC within an Eulerian framework, the computational domains required to analyze
118 cyclone energetics can become quite large (Black and Pezza, 2013, e.g.), potentially
119 incorporating other atmospheric features (e.g., migratory anticyclones, upper-level
120 troughs, large convective systems), which limits the interpretation of results.

121 In this context, Michaelides et al. (1999) developed a Semi-Lagrangian framework
122 for analyzing cyclone energetics, where the computational domain is movable. This
123 approach minimizes the impact of neighboring circulations on the system’s energy
124 dynamics by employing a regional computational domain that follows the cyclone
125 track. In the present study, we present a LEC climatology applied to cyclonic sys-
126 tems using such a framework, as well as the first LEC climatology (considering both
127 Eulerian and Semi-Lagrangian frameworks) for cyclones in the Southwestern Atlantic.
128 Over 6,700 cyclones with genesis in the Southwestern Atlantic region are analyzed,
129 and their energy cycles are divided into distinct life cycle phases: incipient, intensifica-
130 tion, mature, and decay. This division allows for a more comprehensive understanding
131 of the energy flows and the dynamical mechanisms acting across each phase of the
132 cyclone’s life cycle.

133 2 Materials and methods

134 2.1 Data

135 For both the cyclone tracking and energetics computations, we employed the fifth-
136 generation reanalysis from the European Centre for Medium-Range Weather Forecasts
137 (ECMWF), known as ERA5 (Hersbach et al., 2020). ERA5 data are produced using
138 the Integrated Forecasting System (IFS) Cy41r2, which assimilates a wide range of
139 observations, including satellite and in-situ measurements. It provides global coverage
140 of numerous atmospheric, ocean-wave, and land-surface parameters at a horizontal
141 resolution of approximately 31 km (0.25°). ~~In this study, we utilized data from Data~~
142 ~~were requested at~~ all 37 pressure levels, ranging from 1000 hPa to 1 hPa, with a
143 temporal interval of three hours; ~~the verified toolkit analysis used the 32 levels from~~
144 ~~1000 hPa to 10 hPa.~~

145 The cyclone tracks used in this study were obtained from the "Atlantic Extratrop-
146 ical Cyclone Tracks Database" (Gramscianinov et al., 2020). This database tracks the
147 central relative vorticity of cyclones at the 850 hPa level (ζ_{850}) using the TRACK algo-
148 rithm (Hodges, 1994, 1995) and the method outlined by Hoskins and Hodges (2002).
149 This algorithm has been employed in previous studies to assess cyclone climatologies
150 in the South Atlantic region (Gramscianinov et al., 2019, 2020; de Souza et al., 2024).
151 The database covers the period from 1979 to 2020 and spans the entire Atlantic Ocean
152 within the spatial domain of 15°S–55°S and 75°W–20°E. The use of ERA5, instead of
153 other reanalysis datasets, is justified by its higher spatial resolution, which improves
154 cyclone detection in regions with complex orography and temperature gradients, such
155 as the SESA region (Gramscianinov et al., 2020). A detailed description of the tracking
156 methodology can be found in Gramscianinov et al. (2020).

157 The cyclone tracking procedure employed multiple criteria to ensure accurate
158 cyclone detection. These criteria required cyclones to have a duration of at least
159 24 hours and a minimum displacement of 1000 km, consistent with previous South

Atlantic cyclone climatologies (Sinclair, 1995; Gramscianinov et al., 2019). Systems that spent over 80% of their life cycle over continental regions were excluded from the analysis to avoid counting thermal lows and lee troughs (e.g. Crespo et al., 2021). Notably, although the TRACK algorithm’s calibration and sensitivity are focused on extratropical cyclones — the majority of systems in this region (Marrafon et al., 2022) — the methodology does not explicitly exclude subtropical or tropical cyclones.

To focus on the Southwestern Atlantic Ocean, only cyclones with genesis near the South American coast were considered, specifically those originating within the eastern Argentina (ARG), La Plata River basin (LA-PLATA), and southeast Brazil (SE-BR) genesis regions (Gramscianinov et al., 2019; de Souza et al., 2024) (Figure 1). After applying these selection criteria, the database initially comprised a total of 7931 cyclones. The spatial distribution of these genesis regions and the cyclone track density for all selected systems are presented in Figure 1. Noteworthy, in traditional cyclone studies, cyclone density is often presented for a single time step or normalized by individual tracks (e.g., Hoskins and Hodges, 2005; Gramscianinov et al., 2019). In contrast, Figure 1 displays cyclone density computed across all time steps within a 5° spherical cap. Cyclones may form within the cap, move into it from adjacent regions, remain for several time steps, or pass through it entirely. Each of these time-step occurrences is counted, meaning that the density values in the figure reflect multiple temporal passages rather than unique storm events.

Among the original 7,931 cyclones, 4,445 originated in ARG (56.0%), 1870 in LA-PLATA (23.6%), and 1616 in SE-BR (20.4%). However, due to data availability issues during the download process from the ERA5 database, not all cyclones could be considered in this study, resulting in a reduced total of 6,789 cyclones. These issues were related to the ERA5 API instability during the second half of 2023 and the first half of 2024, when ECMWF was migrating its data infrastructure and updating its API services. As a result, slow transfers and repeated interruptions caused incomplete or failed downloads, despite multiple attempts. This led to some cyclones being excluded from the analysis. The remaining dataset still includes a sufficiently large number of systems for robust analysis and generalization of the results. Of the 6,789 cyclones, 4,015 originated in ARG (59.1%), 1,288 in LA-PLATA (18.9%), and 1,486 in SE-BR (21.8%). The reduction corresponds to a 14.4% decrease in the number of cyclones, with region-specific reductions of 9.7% for ARG, 31.1% for LA-PLATA, and 8.0% for SE-BR.

The corrected production rerun validated 3,820 of the 6,789 archived cyclones and 15,829 lifecycle rows. Corrected results in this manuscript therefore use the validated 3,820-cyclone population. Comparisons with the full published population combine the toolkit correction with this population change; a correction-only paired control uses the same 3,820 cyclones on both sides.

2.2 Lorenz Energy Cycle Computation

For computing the LEC, we used the the open-source Python application LorenzCycleToolKit (de Souza et al., 2024). The energy budget equations for the zonal and eddy forms of APE and kinetic energy are then expressed as:

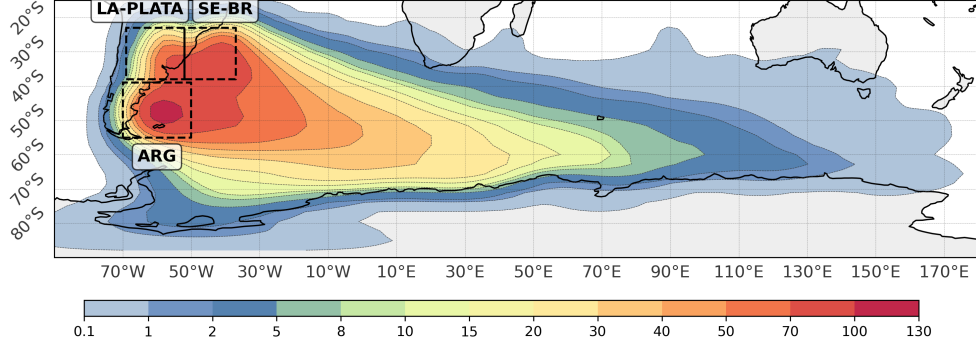


Fig. 1 Cyclone track density for all systems analyzed in this study, highlighting the cyclogenesis regions near the South American coast (ARG, LA-PLATA, and SE-BR). The track density unit is cyclonic centers per 10^6 km^2 per month.

$$\frac{\partial A_Z}{\partial t} = BA_Z - C_Z - C_A + G_Z \quad (1)$$

$$\frac{\partial A_E}{\partial t} = BA_E - C_E + C_A + G_E \quad (2)$$

$$\frac{\partial K_Z}{\partial t} = BK_Z + C_Z - C_K + B\Phi_Z + D_Z \quad (3)$$

$$\frac{\partial K_E}{\partial t} = BK_E + C_E - C_K + B\Phi_E + D_E \quad (4)$$

In these equations, the atmospheric energy reservoirs are separated into zonal and eddy components of available potential energy (A_Z , A_E) and kinetic energy (K_Z , K_E). Energy conversions between reservoirs are represented by the following terms: C_A , the conversion between zonal and eddy available potential energy ($A_Z \leftrightarrow A_E$); C_E , the conversion from eddy available potential energy to eddy kinetic energy ($A_E \leftrightarrow K_E$); C_K , the barotropic conversion from eddy kinetic energy to zonal kinetic energy ($K_E \leftrightarrow K_Z$); and C_Z , the conversion from zonal available potential energy to zonal kinetic energy ($A_Z \leftrightarrow K_Z$). Positive C_K transfers energy from K_E to K_Z , whereas negative C_K feeds K_E . The generation of available potential energy (G) and dissipation of signed frictional contributions to kinetic energy (D) are indicated by subscripts Z and E, such that G_Z and G_E represent the generation of zonal and eddy available potential energy, while D_Z and D_E denote the dissipation of zonal and eddy kinetic energy, respectively; negative D denotes kinetic-energy loss. The boundary flux terms, BA_Z , BA_E , BK_Z , and BK_E , represent the import/export of available potential energy and kinetic energy across the computational domain boundaries. Additionally, the boundary pressure-work terms $B\Phi_Z$ and $B\Phi_E$ appear alongside C_Z and C_E , respectively, as both arise from deriving the kinetic energy balances. According to, these terms are challenging to interpret physically, but indicate processes generating kinetic energy resulting from work performed at the boundaries of the computational domain.

222 either supply or remove kinetic energy. The complete mathematical formulation of
 223 these terms can be found in the Appendix.

224 In these formulations, a global reference state is used to define the APE, meaning
 225 that the computed APE represents the contribution of local APE to global energetics
 226 (Smith, 1980). As a result, it does not account for regional variability, which can
 227 introduce bias in the APE computation (see Novak and Tailleux (2018) for a thorough
 228 discussion). Despite the limitations of this methodology, the formulations by Muench
 229 (1965) and Brennan and Vincent (1980) still provide a valuable framework due to their
 230 simplicity, robustness, and ability to yield consistent, interpretable results, enabling
 231 direct analysis of the dynamic processes driving cyclones development (Dias Pinto
 232 and Rocha, 2011; de Souza et al., 2025, e.g.). Moreover, while using a local APE
 233 definition (Novak and Tailleux, 2018, e.g.) offers more precise results, it often comes
 234 with higher computational costs and does not provide the same level of direct, practical
 235 interpretability.

236 Following the methodologies of Brennan and Vincent (1980); Michaelides (1987);
 237 Veiga et al. (2008); Pezza et al. (2010); Dias Pinto and Rocha (2011), the program
 238 was set to compute the dissipation and generation terms as residuals, defined as:

$$RK_Z = B\Phi_Z + D_Z + \epsilon_{KZ} \quad (5)$$

$$RK_E = B\Phi_E + D_E + \epsilon_{KE} \quad (6)$$

$$RG_Z = G_Z + \epsilon_{GZ} \quad (7)$$

$$RG_E = G_E + \epsilon_{GE} \quad (8)$$

239 Where Here, RK_Z and RK_E are the residual dissipation terms (zonal and eddy,
 240 respectively), composite kinetic-energy residuals: they combine boundary pressure
 241 work, signed frictional dissipation, unresolved-scale transfers, and accumulated numer-
 242 ical error. Likewise, RG_Z and RG_E are the residual generation terms, $B\Phi_Z$ and $B\Phi_E$
 243 represent the boundary pressure work terms, and ϵ accounts for numerical errors in
 244 the computational procedures combine generation with numerical error. These resid-
 245 uals are budget-closure terms and must not be interpreted as direct measurements
 246 of a single physical generation or dissipation process. The set of budget equations
 247 becomes, then:

$$\frac{\partial A_Z}{\partial t} = -C_A - C_Z + RG_Z + BA_Z \quad (9)$$

$$\frac{\partial A_E}{\partial t} = C_A - C_E + RG_E + BA_E \quad (10)$$

$$\frac{\partial K_Z}{\partial t} = C_K + C_Z + BK_Z + RK_Z \quad (11)$$

$$\frac{\partial K_E}{\partial t} = -C_K + C_E + BK_E + RK_E \quad (12)$$

248 A complete depiction of the LEC, with arrows indicating positive energy fluxes,
 249 is presented in Figure 2. This figure also highlights three primary energetic pathways
 250 (baroclinic instability, barotropic conversion, and latent heat release) which will be
 251 discussed throughout the text.

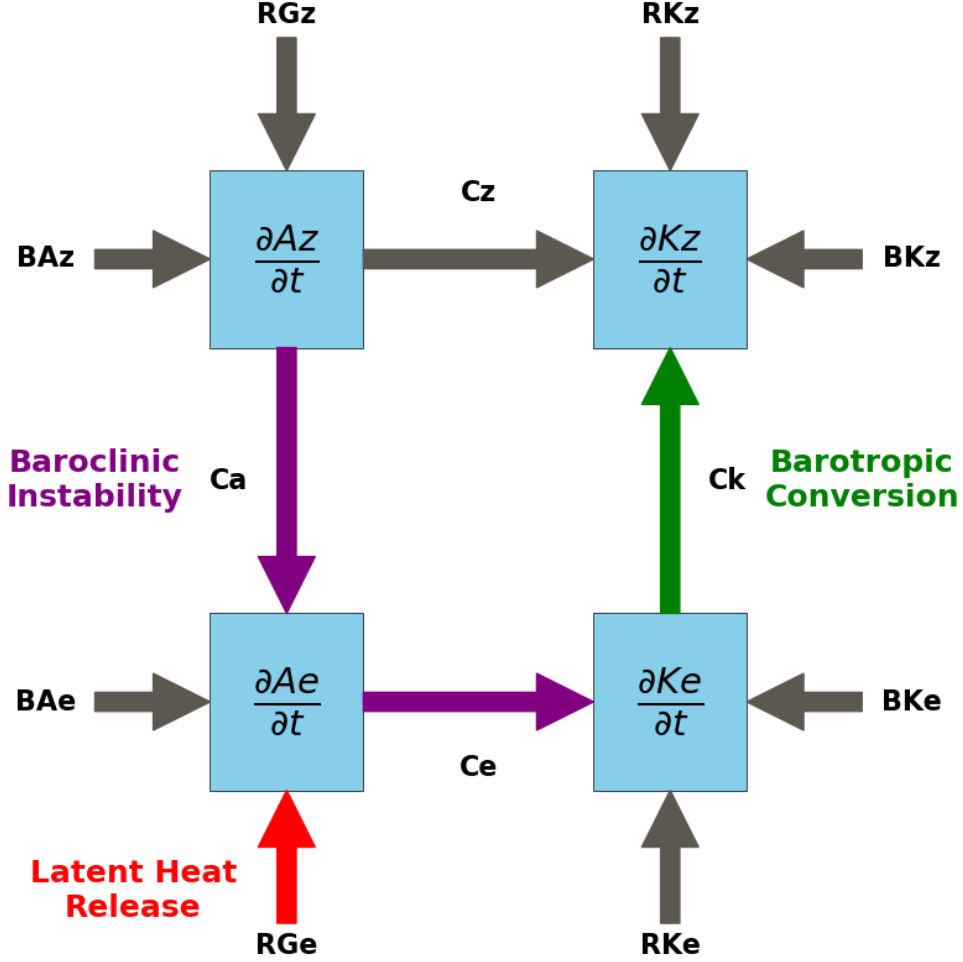


Fig. 2 Representation of the Lorenz Energy Cycle (LEC), with arrows denoting positive energy fluxes. The figure also indicates three main energetic pathways: baroclinic instability (purple), barotropic conversion (green), and latent heat release (red), which will be addressed in detail in the following sections.

252 Here, we used a Semi-Lagrangian framework (Michaelides et al., 1999), aiming to
 253 minimize interactions with other non-related circulations while capturing the main
 254 structure of the cyclone. A $15^\circ \times 15^\circ$ computational domain centered at the cyclone's

central position, obtained from the TRACK database, was created for each time step. Given the impracticality of manually selecting an appropriate domain size for each cyclone in the dataset, a fixed size was employed. The choice of a $15^\circ \times 15^\circ$ computational domain is justified as it is large enough to capture the effective radius of most cyclonic systems (Rudeva and Gulev, 2007). The effective cyclone radius is defined as a measure of cyclone size, determined by establishing a coordinate system centered on the cyclone and measuring the distance at which the radial pressure gradient first falls to zero. This domain size also is large enough for accommodating the cyclone’s synoptic structure (e.g. Gramscianinov et al., 2019).

2.3 Analysis Methods

A key challenge in computing the LEC for large datasets lies in the variable life durations of cyclones. Averaging energy values across an entire life cycle can obscure distinct dynamical processes tied to different stages of cyclone development, while technical limitations exist in dissecting the life cycle into distinct periods. For example, Black and Pezza (2013) averaged cyclone energetics over periods of 48 hours before explosive cyclogenesis, during explosive deepening, and for 24 and 72 hours after it. However, cyclone life cycles can range from less than 24 hours to over 10 days (Trigo, 2006; Reboita et al., 2010; Gramscianinov et al., 2019), presenting significant physical limitations to such approaches.

To address these issues, we employed Cyclophaser, an open-source Python package designed to detect cyclone life cycle phases (de Souza et al., 2025). This program uses time series of relative vorticity at the system’s central position and its first derivative to identify intensification, mature, and decay phases based on peaks and valleys in the vorticity and its derivative. In this study, we adopted the same life cycle detection approach as de Souza et al. (2024). After computing the LEC for each system, Cyclophaser was used to identify the life cycle phases for all analyzed cyclones, followed by the computation of mean LEC values for each phase. This approach facilitates the investigation of dynamical mechanisms specific to each development phase. For example, distinct energy fluxes are expected to dominate during the intensification and decay phases, particularly near the cyclone center. As shown by de Souza et al. (2024), approximately 60% of the analyzed cyclones follow a classical life cycle, consisting of incipient, intensification, mature, and decay phases, while over 95% display at least one intensification phase. A comprehensive description of the detection methodology, as well as spatial distributions and phase-specific statistics such as duration, displacement, speed, and intensity, can be found in de Souza et al. (2024).

To understand the dominant LEC variability patterns within the TRACK dataset, we performed an Empirical Orthogonal Function (EOF) analysis (Fukuoka, 1951; Lorenz, 1956) using the Python open-source library pyEOF (Zheng, 2021). This analysis reduces the dimensionality of the dataset, retaining the most significant variability, and is particularly useful for uncovering underlying structures and patterns in complex environmental data. Typically, each EOF represents a spatial pattern, while the associated time series, or principal component (PC), describes the temporal evolution of that pattern. However, in this study, each EOF reflects a mode of variability derived from all LEC terms, using their average values computed separately for each cyclone

life cycle phase (incipient, intensification, mature, and decay). Consequently, the PCs represent the variation across individual cyclones rather than temporal evolution, and therefore are not displayed individually, as they do not depict any meaningful temporal or spatial feature. Performing EOF analyses separately for each life cycle phase enabled a detailed investigation of the variability in cyclone energetics across distinct stages of cyclone development, as well as throughout the entire energy cycle.

To associate cyclones with predominant EOFs, we first project each cyclone’s energetics onto the PCs obtained from an EOF analysis. This allows us to represent each system in terms of its contribution to the dominant modes of variability. Next, we classify cyclones into EOF(+) and EOF(-) groups, identifying systems where at least one PC reaches extreme values. Specifically, EOF(+) corresponds to cyclones where at least one PC exceeds the 90th percentile (q90), indicating a strong positive projection onto a specific mode. Meanwhile, EOF(-) are cyclones for which at least one PC is below the 10th percentile (q10), indicating strong negative projection onto a specific mode. For each cyclone meeting these criteria, we determine its predominant EOF as the one corresponding to the PC with the highest absolute value. This ensures that each system is classified based on the mode that most strongly characterizes its structure and dynamics.

To determine the LEC patterns of the most intense systems, we first selected only the cyclones in which the maximum central vorticity at 850 hPa exceeded the 90th percentile of the dataset. Subsequently, we applied the K-Means algorithm to the PCs of these intense cyclones to investigate whether distinct energetic characteristics could be identified among them. This clustering analysis was not performed on all cyclones, as sensitivity tests indicated that applying it to the full dataset resulted in groups that differed mainly in intensity rather than exhibiting distinct energetic behaviors (not shown); thus, intensity-based selection was necessary.

K-Means is an unsupervised learning algorithm that partitions data into K clusters by minimizing intra-cluster variance while maximizing separation between groups based on sample similarity (MacQueen et al., 1967; Hartigan and Wong, 1979). The optimal number of clusters was determined using the Elbow Method, which identifies the most suitable number of clusters by plotting the within-cluster sum of squares for different values of K and locating the “elbow point,” where the rate of decrease slows, indicating a balance between model complexity and performance. For this analysis, we employed the K-Means implementation from the `scikit-learn` Python package (Hao and Ho, 2019) and used the `yellowbrick` package for the Elbow Method (Bengfort and Bilbro, 2019).

3 Results

3.1 Climatological Features

The exploratory statistical metrics for all LEC terms, presented as averages across the entire cyclone lifecycle, are shown in Table 1, while their probability density functions (PDFs) are provided in Figure 3. Notably, all terms related to the zonal jets (K_Z , BK_Z , and $\frac{\partial K_Z}{\partial t}$) frequently exhibit statistical metrics one order of magnitude higher

than their eddy counterparts within the same group (K_E , BK_E , and $\frac{\partial K_E}{\partial t}$), indicating significantly greater energy amount concentrated on the jet streams. Although the energy budgets were computed with the generation and dissipation terms calculated as residuals, the generation terms can be directly computed using ERA5 data. Therefore, their probability density functions (PDFs) are also shown for a more physically based analysis (Figure 3e). Furthermore, the comparison between the directly computed generation values and those estimated by residual computation provides a useful metric for evaluating the accumulation of errors in LEC computation procedure. The dissipation terms were not directly computed in the present study.

All the energy terms (K_z , A_z , A_e , and K_e , Figure 3a) exhibit right-skewed distributions. Among these, A_z shows the largest variability, as reflected in its long-tail distribution, standard deviation (std), interquartile range (IQR), and range (defined as the difference between maximum and minimum values), compared to A_E and K_E , peaking at approximately $4 \times 10^5 J m^{-2}$. Conversely, A_E presents the lowest variability, peaking near $1 \times 10^5 J m^{-2}$, closer to its mean value. The kinetic energy terms, K_Z and K_E , display similar distributions, with K_Z peaking near $20 \times 10^5 J m^{-2}$ and K_E peaking near $2 \times 10^5 J m^{-2}$.

Table 1 Summary Statistics of Lorenz Energetics Components, computed as averages across the entire cyclone lifecycle: mean, median, standard deviation (std), 25th percentile (Q25), 75th percentile (Q75), interquartile range (IQR), and range. The IQR measures the range within which the central 50% of the values fall, while the range is computed as the difference between the maximum and minimum values. The units for the energy terms (A_Z , A_E , K_Z , and K_E) are $10^5 J m^{-2}$ and $W m^{-2}$ for the remaining terms.

Term	Mean	Median	Std Dev	Q25	Q75	IQR	Range
A_z	5.55	4.68	3.71	2.77	7.49	4.72	36.46
A_e	1.62	1.24	1.20	0.77	2.09	1.32	11.07
K_z	28.55	26.19	15.10	17.34	37.12	19.78	114.55
K_e	3.70	2.95	2.62	1.92	4.63	2.71	24.81
C_z	0.52	0.38	4.85	-1.96	2.92	4.89	86.70
C_a	0.93	0.46	1.57	0.02	1.44	1.41	19.78
C_k	-3.56	-1.63	9.51	-5.77	0.47	6.24	206.43
C_e	3.84	2.47	4.92	0.63	5.84	5.21	54.74
BA_z	3.54	2.03	7.67	-0.67	6.67	7.34	175.05
BA_e	0.58	0.16	4.40	-1.13	2.02	3.15	75.72
BK_z	-7.93	-5.53	31.75	-23.75	7.83	31.58	445.63
BK_e	0.47	0.08	8.59	-2.73	3.22	5.95	146.87
$B\Phi_Z$	63.87	49.25	128.05	-9.12	132.35	141.47	1419.08
$B\Phi_E$	39.66	31.19	126.50	-28.84	109.47	138.32	1353.99
G_z	-0.31	-0.17	2.66	-1.39	0.81	2.21	68.74
G_e	1.17	0.56	3.14	-0.26	2.21	2.46	58.39
$\frac{\partial A_z}{\partial t}$	-0.07	-0.14	4.46	-2.04	1.68	3.71	106.39
$\frac{\partial A_e}{\partial t}$	-0.23	-0.10	1.92	-0.85	0.51	1.35	48.19
$\frac{\partial K_z}{\partial t}$	1.59	0.75	13.38	-5.33	8.15	13.48	251.24
$\frac{\partial K_e}{\partial t}$	0.28	0.12	3.18	-1.13	1.58	2.71	61.50
RG_z	-2.17	-1.00	7.60	-5.22	1.56	6.78	220.06
RK_z	12.56	8.69	40.49	-8.68	32.71	41.39	561.24
RGe	2.10	1.20	5.15	-0.40	4.21	4.61	97.25
RKe	-7.59	-4.63	10.62	-11.48	-1.04	10.43	138.42

For the conversion terms, The corrected conversion statistics retain the broad variability of C_Z , C_K , and C_E exhibit high variability, in contrast to C_A . The former terms display long-tail distributions, peaking near zero, -1 , and 2 W m^{-2} , respectively, while, which are numerically unchanged by the toolkit correction. The affected terms change materially: in the correction-only paired control, the pooled median C_A presents a narrow distribution with a sharp peak near its mean value, close to zero (Figure 3b). The C_Z probability density function (PDF), along with the mean and quantile values, and its mean value being close to the median, indicate an overall tendency for both $A_Z \rightarrow K_Z$ and $K_Z \rightarrow A_Z$ conversions. However, the positive mean and median values suggest a predominant $A_Z \rightarrow K_Z$ pathway. For C_A and C_E , the metrics presented in Table 1 indicate a consistent tendency for positive conversions, corresponding to $A_Z \rightarrow A_E \rightarrow K_E$ (baroclinic chain), but with more modest $A_Z \rightarrow A_E$ conversions than $A_E \rightarrow K_E$. In the case of the increases from 0.50 to 1.54 W m^{-2} , while the pooled median C_K term (barotropic conversion term), the metrics indicate a predominant negative conversion, i.e., shifts from -1.91 to -0.03 W m^{-2} . Thus, the corrected climatology shows a substantially stronger $A_Z \rightarrow A_E$ conversion and a barotropic conversion whose typical value is close to neutral, with both $K_Z \rightarrow K_E$, although the positive Q75 value suggests the occurrence of events where and $K_E \rightarrow K_Z$ conversions also take place. The published characterization of a predominantly strong $K_Z \rightarrow K_E$ pathway is not supported by the corrected calculation.

For the lateral boundary terms, BA_E and BK_Z exhibit similar distributions, as do BA_Z and BK_E (Figure 3c). The terms BA_E and BK_E display narrow distributions, peaking near zero, whereas BA_Z and BK_Z show nearly symmetrical long-tail distributions. Although for all four terms, both positive (influx) and negative (outflux) energy fluxes occur, as indicated by their negative Q25 and positive Q75 values, the peak near zero for BA_E and BK_Z suggests that boundary energy fluxes are typically low in most cases. In contrast, for BA_Z , the peak near 2 W m^{-2} indicates an overall tendency for energy influx, while for BK_E , the peak near -2 W m^{-2} points to a predominant tendency for energy outflux. Meanwhile, the boundary pressure-work terms ($B\Phi_Z$ and $B\Phi_E$) exhibit significantly higher values compared to the other boundary terms, with both presenting nearly symmetric distributions (Figure 3d). These terms show tail distributions exceeding $\pm 200 \text{ W m}^{-2}$, indicating substantial variability. The physical interpretation of these terms is discussed in Section 4.1 retain their published distributions because these terms are numerically unchanged by the toolkit correction. The boundary pressure-work terms do not. In the corrected phase statistics, mean $B\Phi_Z$ values are approximately 8.4 – 10.2 W m^{-2} and mean $B\Phi_E$ values are approximately -3.0 to -1.0 W m^{-2} , far smaller than the published values. The corrected calculation therefore does not support the published description of anomalously large pressure-work terms.

Overall, both directly computed generation terms (G_Z and G_E) present nearly symmetrical distributions centered near zero (Figure 3e), suggesting that in most cases, either the zonally averaged diabatic heating (G_Z) and the zonal deviations (G_E) are low, or the atmospheric stability is high. However, there are instances where this behavior is amplified ($G < 0$) or reversed ($G > 0$). In contrast, the generation residual terms (are numerically unchanged by the toolkit correction. The residual

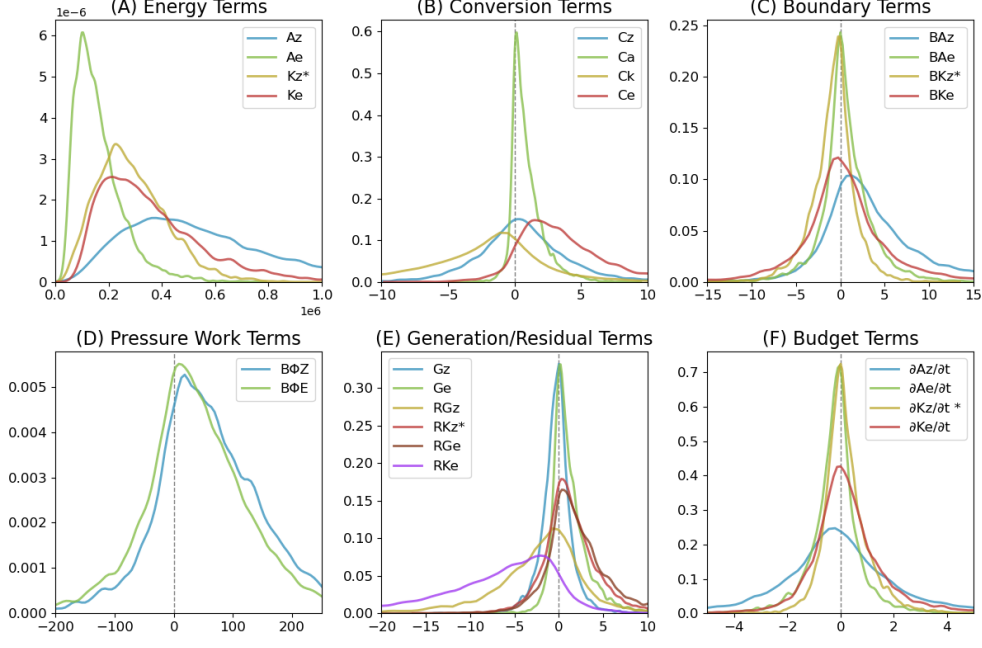


Fig. 3 Probability density function (PDF) plots for distinct Lorenz Energetics terms, computed as averages across the entire cyclone lifecycle: (A) energy, (B) conversion, (C) boundary, (D) pressure work, (E) generation and residual, and (F) budget terms. The x-axis represents the energy values, while the y-axis represents density counts. The units for the energy terms (A_Z , A_E , K_Z , and K_E) are in $10^5 J m^{-2}$, and in $W m^{-2}$ for the remaining terms. Terms related to K_Z are divided by 10 (*) to account for their order of magnitude difference, which would otherwise distort the visualization.

terms RG_Z and RG_E) display distinct characteristics. The RG_Z term is left-skewed, with a long-tail distribution and higher variability compared to G_Z , as indicated by its lower Q25 and higher Q75 and IQR values (Table 1). Additionally, RG_Z exhibits more negative mean and median values than G_Z , suggesting a bias toward processes such as diabatic cooling at lower latitudes and/or heat distribution mechanisms that reduce zonal temperature gradients, such as cyclonic activity. On the other hand, RG_E is more symmetrical than RG_Z , though it still shows higher variability compared to G_E , as evidenced by its lower Q25 and higher Q75 and IQR values. Moreover, RG_E exhibits higher mean and median values than G_E , indicating a bias toward overestimating the eddy effects on latent heat release and meridional temperature gradients. Therefore, the residuals suggest that the computed LEC increases the contribution of eddy motions to the atmospheric energy cycle through enhanced meridional, however, inherit the C_A correction through budget closure. Their differences from the direct generation terms cannot be assigned uniquely to diabatic heating, meridional heat transport, reducing zonal temperature gradients and enhancing convective activity within cyclone frontal structures.

Similar to the generation-residual terms, each dissipation-residual term displays distinct distributions (Figure 3e). The PDF for RK_Z mirrors that of RG_E , showing

modest variability. Its negative Q25 values indicate instances of net dissipation of K_Z . However, the high positive mean, median, Q75, and IQR values suggest that in most cases, there is a net increase of K_Z related to this term, possibly due to the high $B\Phi_Z$ value. In contrast, RK_E shows high variability with a left-skewed or convective activity, long-tail distribution. Both its Q25 because each residual also contains accumulated numerical error and unresolved contributions.

The composite kinetic residuals RK_Z and RK_E are negative, indicating that dissipation of K_E frequently occurs. RK_E also inherit the C_K correction through budget closure. Their signs describe the net closure contribution but cannot be interpreted as frictional dissipation alone. In particular, the former explanation based on anomalously large $B\Phi_Z$ is no longer supported after the pressure-work correction.

3.2 Mean Energy Values Across Life Cycle Phases

The use of the CycloPhaser program allowed for the application of an objective criterion to dissect the cyclones into distinct life cycle phases, enabling an investigation of their energy cycles across these phases (Figure 4). Our analysis revealed that the mean energy flow directions rarely change across phases, although the magnitude of each term varies. However, the high standard deviation values indicate that energy fluxes frequently deviate from their mean direction.

The A_Z budget term exhibits mean positive values during the incipient phase, which turn negative during the intensification and mature phases, becoming positive again during the decay phase. The mean K_Z budget values are positive throughout all phases except during the mature phase. In contrast, the A_E budget remains negative across the entire lifecycle, while the K_E budget is positive during the incipient and intensification phases but turns negative during the mature and decay phases. However, it is important to note that the large variability observed in these terms indicates that numerous individual cases deviate significantly from these mean behaviors, including instances of opposite signs or substantially greater magnitudes.

Across the different cyclone phases, the mean behavior suggests that both baroclinic and barotropic conversions remain active, continuously providing energy to the eddies, while the adiabatic contribution from the G_E term is variable throughout the lifecycle. There is a marked enhancement of baroclinic conversions from the incipient to intensification phase, which then decrease in magnitude during the corrected mean behavior is dominated by baroclinic conversion. Mean C_A increases from 1.92 W m^{-2} in the incipient phase to 4.12 W m^{-2} during intensification, then decreases to 3.67 and 2.16 W m^{-2} during mature and decay phases. Conversely, barotropic conversion peaks during the mature phase, surprisingly with a higher magnitude than the $A_E \rightarrow K_E$ conversion. The behavior of the baroclinic chain is mirrored by the Mean C_K is 0.33 W m^{-2} during incipience, close to zero during intensification (-0.30 W m^{-2}), and modestly eddy-feeding during maturity and decay (-2.43 and -2.10 W m^{-2}). Barotropic conversion is therefore case-dependent and is not a continuous dominant source for eddies. The unchanged G_E term, which begins with negative values and remains negative during incipience and positive thereafter, peaking during intensification, during the intensification phase, reaches a higher magnitude than the C_A term. The large variability observed in these energy conversion terms

indicates that multiple cyclone configurations likely exist, where either baroclinic or barotropic conversions dominate, or where both pathways coexist with varying intensities. Furthermore, the variability of the G_E term suggests that, in some systems, adiabatic heating plays a significant role in cyclonic development, while in others, diabatic cooling hinders the development. The G_Z term remains negative throughout the entire life cycle, with progressively decreasing values.

Throughout the life cycle, the mean values indicate imports of A_Z , which decrease in the mature phase but rise again during the decay phase. The mean BA_E values exhibit a bimodal behavior, being positive during the incipient and intensification phases but turning negative during the mature and decay phases, though with negligible mean values in the latter. However, the large variability observed suggests that for individual cyclones, imports and exports of A_Z and A_E can differ substantially from these mean patterns, with some systems potentially exhibiting more pronounced or negligible boundary flux contributions.

Across all phases, the RK_Z term stands out with the highest mean values, also displaying the greatest variability, as indicated by the high standard deviation values. This could be influenced by the $B\Phi_Z$ term, as mentioned in the previous section. The BK_Z term also presents high mean values and variability across all phases, suggesting a compensatory effect for the excess energy from the RK_Z term. Lastly, the mean BA_E values are negative across all phases, peaking during the mature phase, indicating significant dissipation in this phase and presenting similar values during the incipient and decay phases. The corrected pressure-work calculation removes the basis for attributing the magnitude of RK_Z to anomalously large $B\Phi_Z$, and neither residual should be read as a direct dissipation estimate.

3.3 Empirical Orthogonal Function Analysis

To better characterize the substantial variability observed in the LEC terms, we performed an Empirical Orthogonal Function (EOF) analysis to identify distinct energetic configurations throughout the cyclone lifecycle. The first EOF accounts for 28.3% of the total variance, followed by the second and third EOFs with 11.0%. In the corrected population, the leading mode explains 27.7% of the variance. The modes pattern-matched to published EOF2, EOF3, and EOF4 explain 10.9%, 10.5%, and 4.4%, respectively. The fourth to eighth EOFs contribute progressively less (8.2%, 7.4%, 5.9%, 5.2%, 4.1%, and 3.8%), with the first four modes explaining 58.47.6%, respectively; the matched EOF4 is ranked fifth in the corrected solution, demonstrating that higher modes can exchange order. The first eight corrected modes explain 81.3% of the total variance and the first eight reaching 81.3%. These results highlight the dominant patterns of variability, with diminishing contributions beyond the initial modes. This study focuses on the first four EOFs. Although EOFs can be interpreted in terms of both positive and negative contributions, we limit our analysis to the positive phase for simplicity. Comparisons below must therefore follow the one-to-one matched loading patterns rather than numerical rank alone.

Each cyclone can be described as a linear combination of the EOFs, with the corresponding principal components (PCs) indicating how strongly each mode contributes

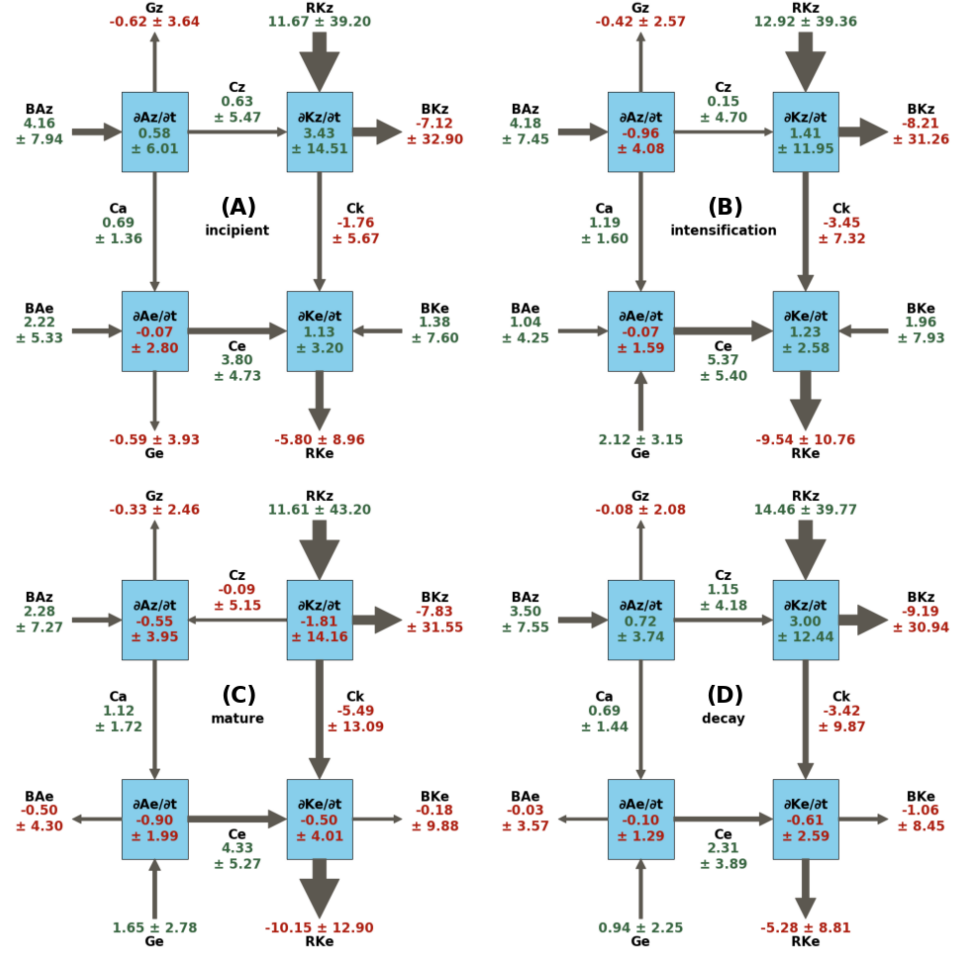


Fig. 4 Mean values and standard deviation of the limited-area Lorenz Energy Cycle (LEC) terms for each life cycle phase : incipient (A), intensification (B), mature (C), and decay (D). Each box represents an energy component ($J m^{-2}$), with arrows showing the direction of energy flow ($W m^{-2}$). The numbers adjacent to the arrows indicate the magnitude and direction of the energy flux, with green indicating positive values and red indicating negative values

to the system's energetic structure. Consequently, the importance of a given EOF varies from system to system. The energy fluxes associated with each EOF represent deviations from the climatological mean (Figure 4), either reinforcing or weakening specific energy pathways. In this way, each EOF reflects the amplification or attenuation of a characteristic energy transfer pattern within the LEC. In section 3.4, we examine cyclones that project strongly onto each EOF individually.

Figure 5 illustrates the Lorenz Energy Cycle (LEC) terms associated with EOF 1. Across all life cycle phases, there is a consistent enhancement of moist baroclinic

During the incipient (Figure 6A) and intensification phases (Figure 6B), EOF 2 displays an overall weakening of the C_A term, in contrast to EOF 1. Additionally, there is a reduction in barotropic conversions, indicating a $K_E \rightarrow K_Z$ tendency, alongside a weakening of K_E imports. This reduction is compensated by an enhancement in A_E imports, which appears to be the primary mechanism driving the increase in the C_E term. In the mature phase (Figure 6C), the signal for A_E imports reverses in sign, and an enhancement of the moist baroclinic chain occurs, accompanied by a slight increase in $K_Z \rightarrow K_E$ conversions and a more pronounced increase in K_E imports. During the decay phase (Figure 6D), both the moist baroclinic chain and K_Z imports show a negative tendency. However, the enhanced barotropic conversions, coupled with a weakening of dissipation (positive tendency for RK_E), result in a slight yet positive tendency in K_E , suggesting a slower dissipation process.

For EOF 3 (Figure 7), the role of the moist baroclinic chain diminishes in eddy development. This is evident from the weakening of the C_E and G_E terms across the incipient (Figure 7A), intensification (Figure 7B), and mature (Figure 7C) phases, although the C_A term shows a slight enhancement during these phases. As a result, we can expect a weaker development for the incipient phase, indicated by the reduced K_E budget. However, during the intensification and mature phases, the dominant source of energy shifts to barotropic conversions. During the decay phase (Figure 7D), the marginal enhancement of the C_E term is driven by a slight increase in the G_E term and, more importantly, by an enhancement of A_E imports. Although this works in conjunction with increased K_E imports, the simultaneous enhancement of $K_E \rightarrow K_Z$ conversions leads to the dissipation of eddies.

During the incipient phase (Figure 8A) of EOF 4, the moist baroclinic chain weakens, as do the K_E imports, while the barotropic conversions are enhanced. As the system progresses to the intensification phase (Figure 8B), the barotropic conversion signal becomes neutral, and the weakening of the baroclinic chain lessens, accompanied by a strengthening of A_E imports. In the mature phase (Figure 8C), both the baroclinic chain and the A_E and K_E imports are slightly enhanced. However, the barotropic conversions reverse, signaling $K_E \rightarrow K_Z$ conversions, indicating a transfer of energy from eddy kinetic energy to zonal kinetic energy. In the decay phase (Figure 8D), the signal of baroclinic conversions returns to its initial weakened state and A_E imports are reduced, while K_E are enhanced. Although the $K_E \rightarrow K_Z$ conversions are enhanced, they are counterbalanced by the K_E imports, leading to a slower eddy dissipation (higher tendency for K_E the budget).

Here we present a summary of the energy pathways that were enhanced for each EOF, indicating the increased contribution to cyclone development (i.e., K_E increases). During EOF1, both the moist baroclinic and barotropic chains are enhanced, with increased imports of K_E during the incipient and intensification phases. For EOF2, cyclone development is characterized by the enhancement of the $BA_E \rightarrow A_E \rightarrow K_E$ chain during the incipient and intensification phases, followed by an increase in K_E imports during the mature phase. For EOF3, during the incipient phase, there is an enhancement of K_E imports, which is followed by an increase in barotropic conversions during the intensification and mature phases. For EOF4, barotropic conversions are enhanced during the incipient phase, with nearly neutral baroclinic and barotropic

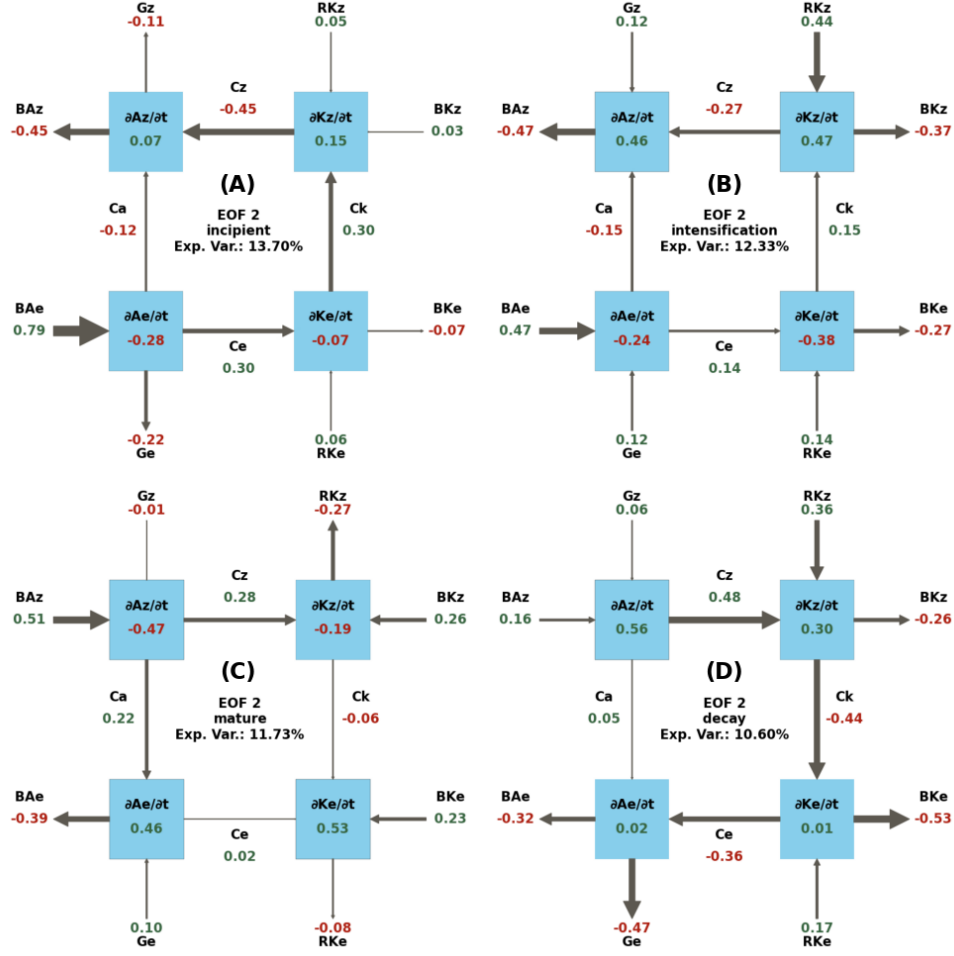


Fig. 6 Lorenz Energy Cycle (LEC) terms for the second Empirical Orthogonal Function (EOF 2) across different life cycle phases: (A) incipient, (B) intensification, (C) mature, and (D) decay.

conversions during intensification, and an enhanced baroclinic chain along with K_E imports during the mature phase. During cyclone decay, EOF1 shows an enhancement in dissipation, EOF2 and EOF4 present an enhancement in exports of K_E , and EOF3 exhibits an increase in $K_E \rightarrow K_Z$ conversions.

3.4 Characteristics of Cyclones Based on EOF Signal

In this section, as the characteristics of the systems associated with each EOF are analyzed, the EOF signal becomes relevant, necessitating a distinction between systems related to positive EOF signals — EOF(+) — and negative EOF signals — EOF(-), respectively. EOF1 accounts for the largest number of cyclones among the first four

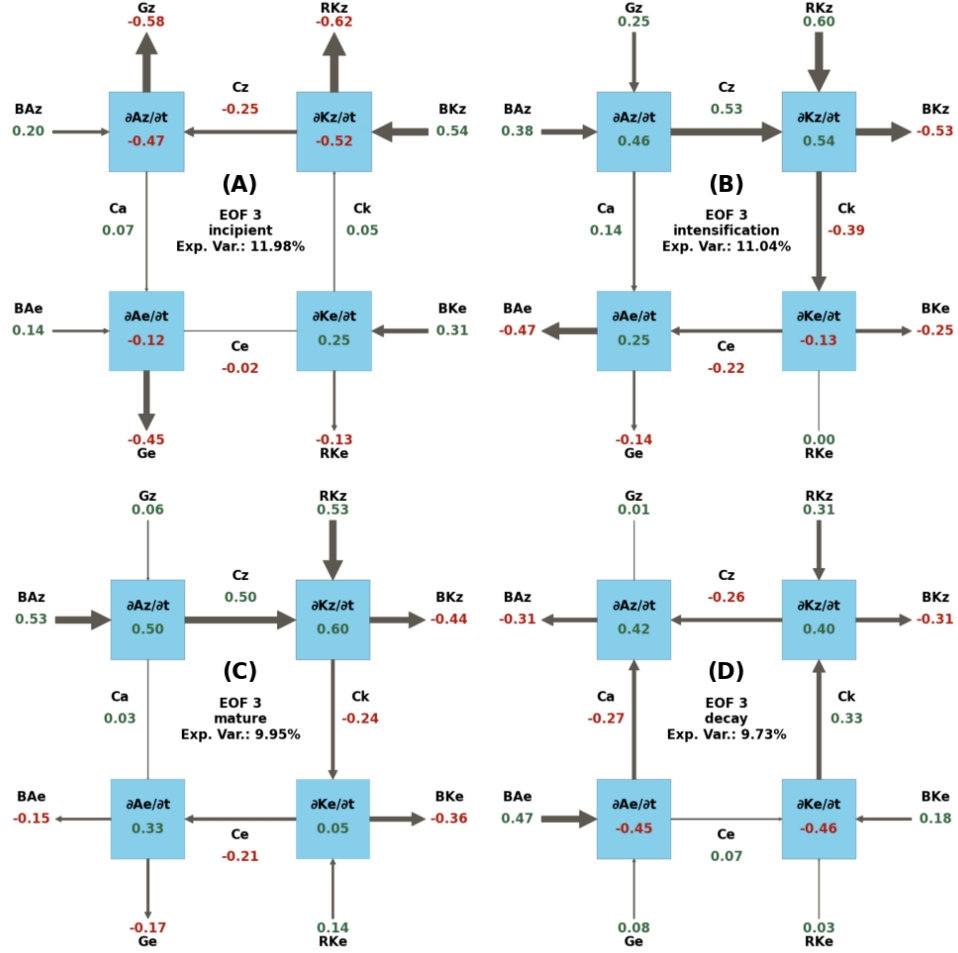


Fig. 7 Lorenz Energy Cycle (LEC) terms for the third Empirical Orthogonal Function (EOF 3) across different life cycle phases: (A) incipient, (B) intensification, (C) mature, and (D) decay.

580 EOFs for both EOF(+) and EOF(-). For EOF(+), it is followed by EOF2, EOF3,
 581 and EOF4, while for EOF(-), it is followed by EOF3, EOF2, and EOF4, respectively
 582 (Figure S1a). The track densities for the cyclones associated with EOF(+), are pre-
 583 sented in Figure 9, while those for EOF(-) are shown in Figure 10. For EOF1(+),
 584 the track density maximum extends from SE-BR and LA-PLATA toward the ARG
 585 region, highlighting the significant contribution of systems originating in the former
 586 regions compared to the latter, as also indicated in Figure 11a. In contrast, for the
 587 remaining EOFs(+), the track density maximum is located near the ARG region, as
 588 most systems originate there. An exception is observed in EOF3(+), where the track

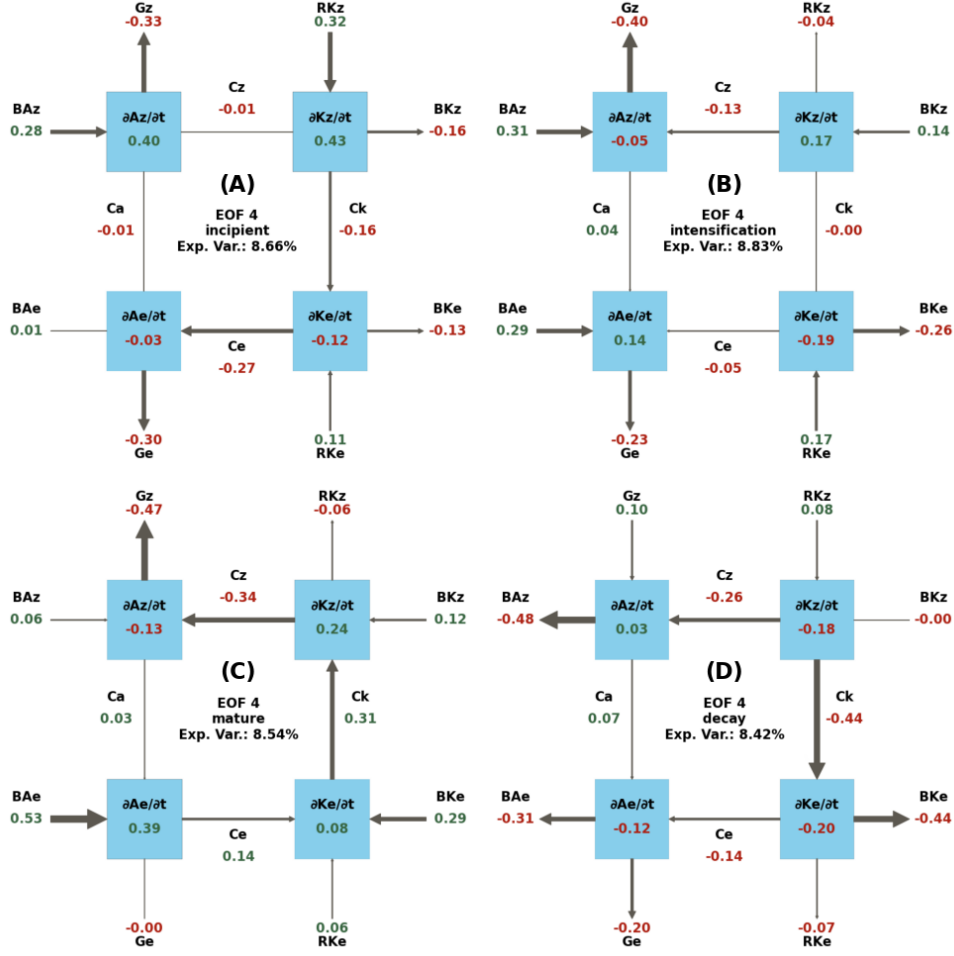


Fig. 8 Lorenz Energy Cycle (LEC) terms for the fourth Empirical Orthogonal Function (EOF 4) across different life cycle phases: (A) incipient, (B) intensification, (C) mature, and (D) decay.

density maximum extends northeastward due to the increased contribution of SE-BR systems to this EOF.

Meanwhile, EOFs(-) exhibit a more variable behavior. EOF1(-) displays two track density maxima: one near SE-BR and another near ARG, as fewer systems originate from LA-PLATA for this EOF (Figure 11c). EOF2(-) is predominantly influenced by ARG systems, whereas EOF3(-) shows an increased contribution from LA-PLATA systems, with a primary track density maximum near ARG and a secondary maximum over the La Plata River mouth. Lastly, EOF4(-) presents a track density maximum between LA-PLATA and SE-BR, reflecting the increased relative contribution of systems from these regions to this EOF.

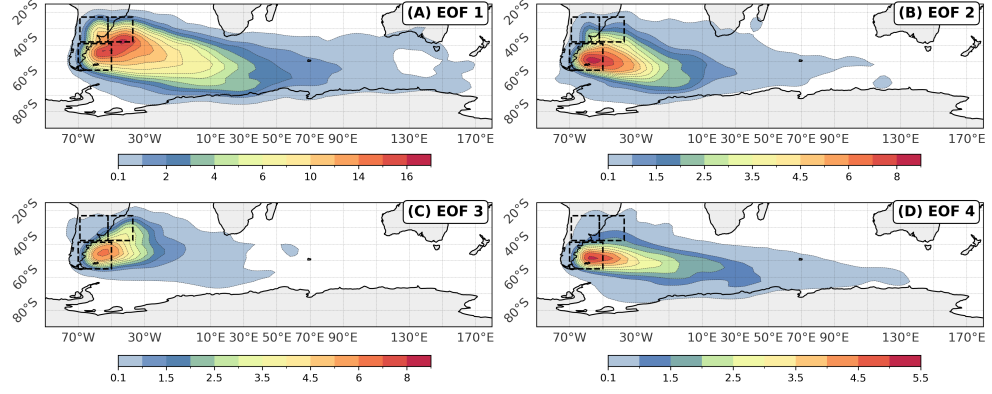


Fig. 9 Cyclone track density for the systems associated with EOFs(+): (A) EOF1, (B) EOF2, (C) EOF3, and (D) EOF4. The track density unit is cyclones per 10^6 km^2 per month. The dashed rectangles mark the genesis regions shown in Figure 1.

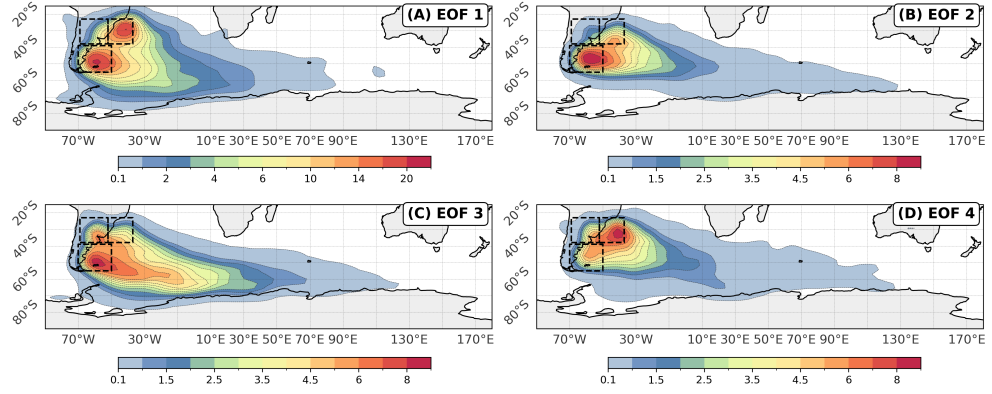


Fig. 10 Cyclone track density for the systems associated to EOFs(-): (A) EOF1, (B) EOF2, (C) EOF3 and (D) EOF4. The track density unit is cyclonic centers per 10^6 km^2 per month. The track density unit is cyclones per 10^6 km^2 per month. The dashed rectangles mark the genesis regions shown in Figure 1.

Despite regional differences in track density maxima, nearly all EOFs exhibit a common pattern of cyclone tracks extending southeastward across the South Atlantic (Figures 9 and 10), with the exception of EOF3(+). Some systems display exceptional mobility, reaching as far as $150^\circ E$. This behavior aligns with the typical lifecycle of cyclonic systems originating near the South American coast, where cyclones generally develop near the coastal region, mature as they propagate southeastward, and decay closer to the Antarctic region (de Souza et al., 2024).

As shown in Figure 11b, EOF1(+) and EOF4(+) systems predominantly originate during JJA, while EOF2(+) and EOF3(+) genesis events are more evenly

distributed across the seasons. For EOFs(-), EOF1(-) systems occur mainly during DJF, whereas EOF4(-) systems are more frequent during DJF and MAM. In contrast, EOF2(-) and EOF3(-) systems are predominantly observed during JJA and SON (Figure 11d). The seasonal distribution of EOF2(+) and EOF3(+) systems aligns with the expected behavior, as ARG systems are well-distributed throughout the year (Gramscianinov et al., 2019; Crespo et al., 2021). However, for the remaining EOFs, the observed behavior suggests that seasonal variations are linked to changes in the systems' energetics.

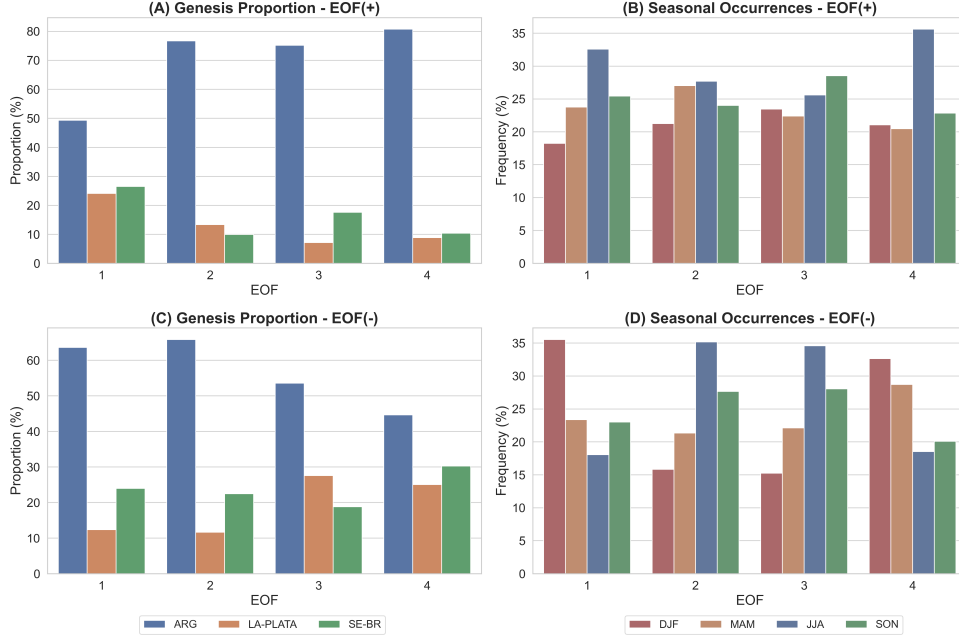


Fig. 11 Genesis proportion and seasonal occurrences of EOFs. (A) Genesis proportion for EOF(+), (B) Seasonal occurrences for EOF(+), (C) Genesis proportion for EOF(-), and (D) Seasonal occurrences for EOF(-). The colors represent different genesis regions (ARG, LA-PLATA, SE-BR) in (A) and (C), while in (B) and (D), the colors correspond to seasonal distributions (DJF, MAM, JJA, SON).

3.5 Groups of Intense Systems

While the previous sections examined the LEC for all cyclonic systems in the South Atlantic — analyzing both the mean behavior and its associated variability through EOF analysis — this section focuses on the energetic groups associated with the most intense cyclones in the dataset. The K-Means algorithm was applied to the first eight PCs of the selected cyclones, identifying five distinct LEC groups, as determined by

Cluster 4 features the weakest energy conversions overall, despite showing nearly doubled values for baroclinic and barotropic terms relative to the full-sample average, while still presenting below-average K_E imports. Across all clusters, the residual term is not strictly proportional to the K_E budget: Cluster 1, for instance, shows the highest RK_E despite having a lower K_E budget than Clusters 2 and 3. This suggests that, although dissipation likely dominates the residual magnitude, contributions from subgrid processes and numerical errors cannot be ruled out. The corrected reconstruction retains five energetically distinct groups, but the published cluster-by-cluster pathway descriptions cannot be carried over without revision because both affected conversion terms enter the clustering and group reconstruction. The composite residuals also cannot be interpreted as direct dissipation estimates.

Figure 13 displays the track density for all clusters. For all clusters, the track density maxima extend from a region near the southeastern South American coast, between 30°S and 50°S , propagating southeastward toward the Antarctic region. Cluster 2 and 3 present tracks more concentrated near the continent, related to their lower mean duration (Figure S2). In contrast, Cluster 1 presents a track maxima spread from near 60°E to near 10°E . Meanwhile, Cluster 4 presents a maxima near the continent and other near Antarctica. While the track density maxima near LA-PLATA and SE-BR are associated with the incipient and intensification phases of these systems, the maxima near Antarctica correspond to their decay phase (de Souza et al., 2024). The relatively small displacement of these systems indicate that they might be related to explosive cyclogenesis and in fact, among explosive cyclogenesis in South America, evidence suggests a great contribution from LA-PLATA (Andrade et al., 2024, 2025), however, further investigation is needed to confirm this.

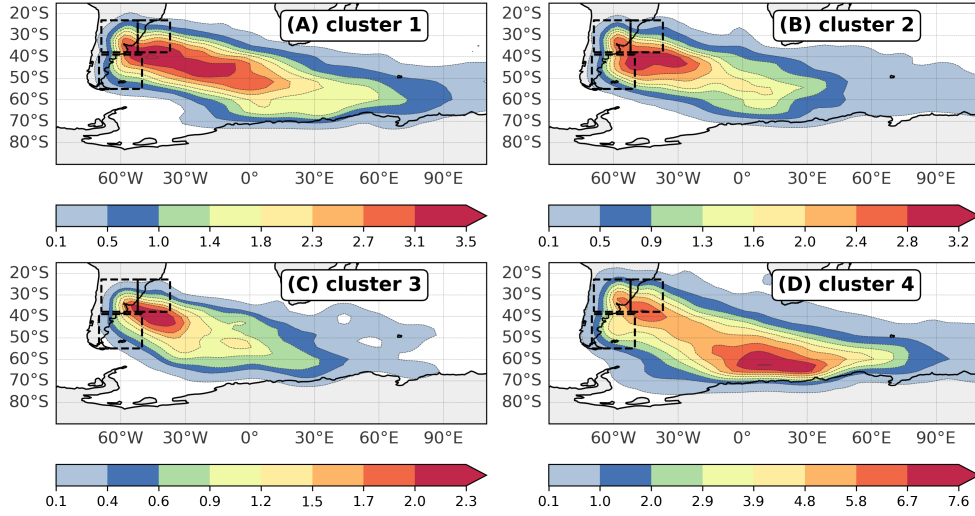


Fig. 13 Cyclone track density for the five clusters (A–E) identified among the most intense cyclones in the dataset. These systems are defined as those with maximum central relative vorticity at 850 hPa exceeding the 90th percentile. The dashed rectangles mark the genesis regions shown in Figure 1.

Cluster 4 has the highest number of systems (Figure 14a), although these are generally less intense (Figure 14b), followed by Clusters 1, 2, and 3. Cluster 3 exhibits the most intense systems on average, while Clusters 1 and 2 show similar mean intensity. Clusters 1 and 2 are more frequent during JJA, while Cluster 3 peaks in SON and Cluster 4 shows no strong seasonality (Figure 14c). In all clusters, the proportion of cyclones originating in the corrected five-group reconstruction, the matched groups contain 647, 266, 229, 206, and 138 systems. Across the full corrected intense subset, 53.3% originate in ARG, 28.7% in LA-PLATA (Figure 14d) exceeds its baseline occurrence in the full dataset (18.9, and 18.0% in SE-BR; JJA is the most frequent season (27.4%), reinforcing the region's role in the development of intense systems, as noted by . The seasonal distributions are also consistent with prior findings for the closely followed by SON (26.1%). LA-PLATA region . The low frequency of cyclones forming in SE-BR is expected, as these tend to be weaker . Although subtropical cyclones are most frequent over SE-BR during DJF , their limited presence in this intense cyclone subset prevents them from significantly influencing the results remains overrepresented relative to its 18.9% share of the full published dataset, but it is not the most frequent absolute genesis region. Group-specific labels and interpretations require care because the corrected clusters are matched by centroid and do not preserve membership.

4 Discussion

4.1 Key Features

The LEC diagrams (Figure 4) indicate that the mean energetic behavior is preserved across distinct life cycle phases. However, the high variability shown in Figure 3 reveals a wide range of behaviors among South Atlantic cyclones. While this section focuses on the mean energy cycle, it is important to recognize that energy budget terms serve primarily as a starting point. The energy conversions and fluxes offer more robust insight into the dynamical processes governing cyclone development. In addition to the phase-specific diagrams, a schematic of the mean Lorenz Energy Cycle for all systems is shown in Figure 15, facilitating comparison across phases and highlighting dominant pathways.

The A_Z term increases during the early phases due to persistent influxes across the boundaries ($BA_Z > 0$), rather than local generation (G_Z). As baroclinic conversions intensify, the A_Z budget becomes negative during the intensification and mature phases, before returning to positive in the decay phase. This evolution suggests that A_Z variations are not driven by latitudinal diabatic heating contrasts. A similar pattern was reported by Dias Pinto and Rocha (2011), although their methodology excluded the incipient phase due to the use of sea-level pressure for cyclone tracking, instead of relative vorticity at 850 hPa (Sinclair, 1994; Hoskins and Hodges, 2002).

For K_Z , a general increase is observed except during the mature phase, when strong barotropic conversion ($K_Z \rightarrow K_E$) coincides with negative C_Z values. This indicates descending motion at warmer latitudes and ascending motion at colder latitudes as the cold front reaches the equatorward side of the cyclone and the warm front reaches the poleward side. The corrected mature-phase mean C_K remains negative, but its

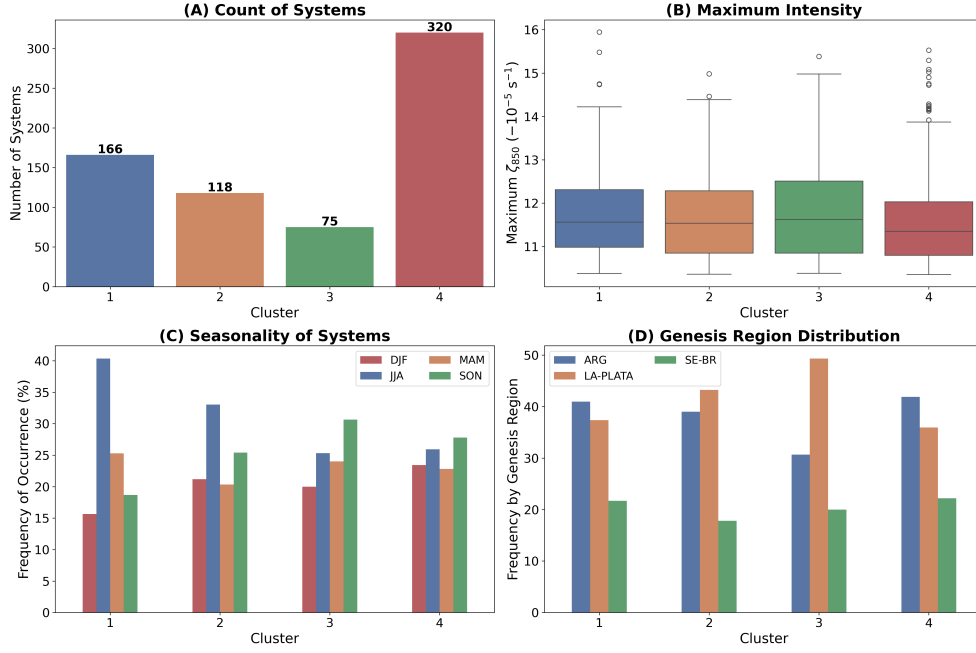


Fig. 14 Characteristics of the five clusters (A–E) identified among the most intense cyclones in the dataset. (A) Number of cyclones per cluster, (B) maximum intensity (defined as maximum central relative vorticity at 850 hPa), (C) seasonal distribution of occurrences, and (D) genesis region count. In (C), colors represent different seasons (DJF, MAM, JJA, SON), while in (D), colors correspond to genesis regions (ARG, LA-PLATA, SE-BR).

magnitude is substantially reduced relative to the published value and is smaller than the corrected C_A . Notably, BK_Z and RK_Z present high magnitudes: the former suggests—indicates export of zonal kinetic energy (e.g., jet stream outflow)—, while the latter aggregates—is a composite closure term that aggregates signed frictional dissipation, boundary pressure work ($B\Phi_Z$), subgrid $B\Phi_Z$, unresolved-scale transfers, and numerical errors.

As the Semi-Lagrangian framework does not account for changes in the background energetics along a moving domain, part of the apparent time evolution in A_Z and K_Z may simply reflect the cyclone’s poleward drift (de Souza et al., 2024). Such drift carries the system into latitudes of larger climatological background APE, implying an increase in A_Z (Novak and Tailleux, 2018; Liu et al., 2024). Because the zonally integrated K_Z reaches its climatological maximum near 60°S (Novak and Tailleux, 2018; Liu et al., 2024), K_Z should decrease for storms that originate over ARG and migrate poleward, but increase for those formed in SE-BR or LA-PLATA. In practice, however, both reservoirs evolve irregularly and non-monotonically, indicating that intrinsic cyclone dynamics, rather than passive domain advection, control the local energetics, corroborating Federer et al. (2025).

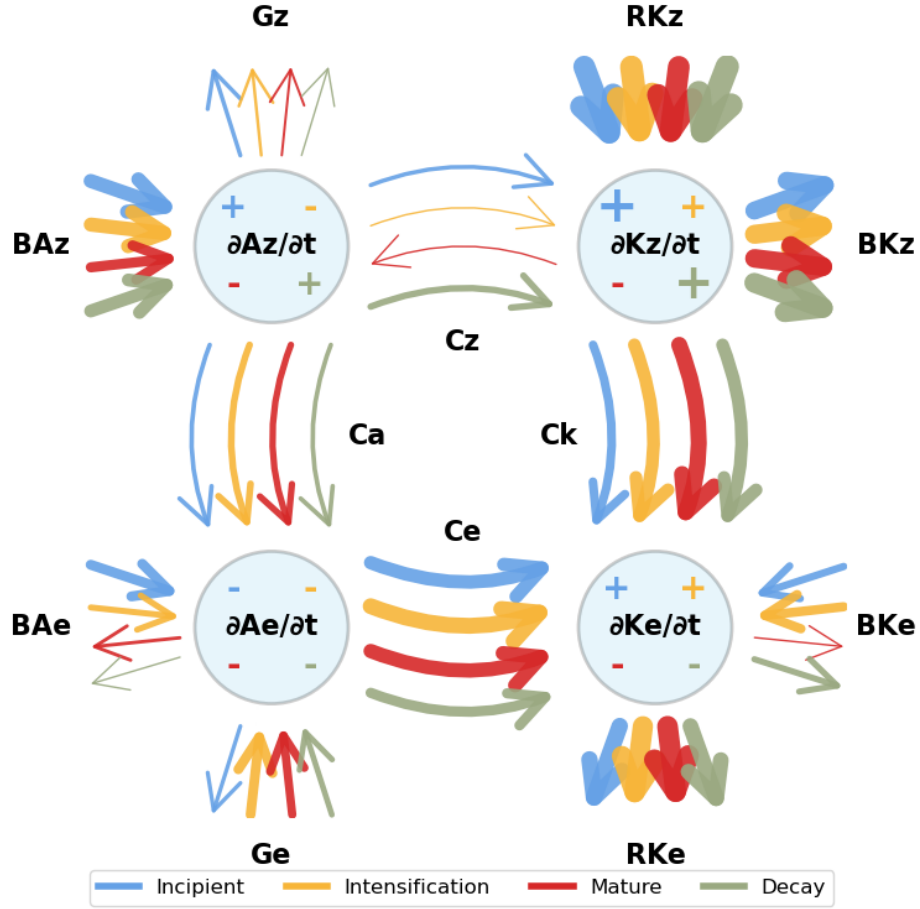


Fig. 15 Schematic representation of the mean Lorenz Energy Cycle (LEC) for all analyzed cyclones in the South Atlantic, combining information from all life cycle phases. Arrows represent energy fluxes between reservoirs, with colors indicating the life cycle phases. The arrow thicknesses and symbol sizes for the balance terms ($\partial/\partial t$) are proportional to the absolute magnitude of each term, as presented in Figure 4.

724 The A_E budget is negative throughout the life cycle, with minimal values during
 725 the incipient and intensification phases. This reflects a highly active and efficient
 726 baroclinic chain. Energy is transferred from A_Z via C_A , and from A_E to K_E via C_E ,
 727 driven by meridional heat transport and vertical motions in frontal zones. During
 728 intensification, G_E becomes positive, associated with convective heating along the cold
 729 front (e.g., Govekar et al., 2011). Simultaneously, radiative heating in the warm sector
 730 due to long-wave absorption from mid- to high-level clouds (Lau and Crane, 1997;
 731 Keshtgar et al., 2023), combined with latent heat release from intense precipitation

during this phase (McErlich et al., 2023), enhances G_E . As the system matures and convection weakens, BA_E reverses (export), and the A_E budget sharply decreases. During decay, with disorganized frontal structures and weak diabatic forcing, A_E returns to near-zero budget values.

The K_E reservoir increases during incipient and intensification phases, ~~as baroclinic, barotropic, and boundary fluxes when~~ baroclinic conversion and boundary imports supply energy. ~~In the mature phase, barotropic conversion peaks, but weakening of baroclinic forcing and reversal of boundary fluxes, along with increased~~ Corrected C_K is weak and of mixed sign in these early phases. During the mature and decay phases, mean C_K is eddy-feeding but remains smaller in magnitude than C_A , while changes in boundary fluxes and the composite RK_E ~~lead to a~~ accompany net energy loss. ~~During the decay phase, diminishing energy inputs and rising exports further reduce K_E . Although~~ Because RK_E includes numerical errors and the $B\Phi_E$ term, the trend suggests dominant dissipative processes, in line with several unresolved contributions, it does not by itself demonstrate dominant physical dissipation.

Both $B\Phi_Z$ and $B\Phi_E$ were ~~directly computed and yielded large values (Table 1, Figure 3d). These terms, representing pressure work by zonal and meridional winds at the domain boundaries, are highly sensitive to small errors in the geopotential field. Despite quality control in ERA5, some inconsistencies may persist, and since the LECTK tool does not apply preprocessing filters, such errors likely propagate into the diagnostics. Future work should examine the physical interpretation of $B\Phi$ terms and the causes of their anomalous magnitudes, as they are seldom reported in literature~~ recomputed with corrected boundary formulations. Their magnitudes collapse relative to the published values, and $B\Phi_E$ changes sign in most paired cyclone phases. The corrected results therefore remove the basis for attributing their published amplitudes to ERA5 geopotential errors or for describing those amplitudes as a physical feature of the cyclones.

4.2 Cyclone Groups

Substantial variability in the dataset, as revealed by the EOF analysis, is largely captured by EOF1(+) and EOF1(-), which together represent nearly 40% of all systems. EOF1(+) is linked to stronger cyclones, while EOF1(-) corresponds to weaker ones (Supplementary Figure S1b), indicating a bi-modal structure in cyclone intensity. This suggests that "day-to-day" systems tend to follow distinct energetic regimes. Traditional cyclone climatologies over the South Atlantic tend to group all cyclones together (Sinclair, 1994; Hoskins and Hodges, 2005; Reboita et al., 2010; Gramscianinov et al., 2019, e.g.). In contrast, the results presented here suggest that these systems exhibit distinct behaviors and may therefore be categorized into separate groups.

These intensity differences manifest in their energy cycles and physical characteristics. Figure 16 presents schematic LECs for the four leading EOF(+) groups. EOF1(+) displays a consistent enhancement across all energy pathways, while EOF1(-) shows a reduction. These differences extend to cyclone mobility: EOF1(+) systems follow the South Atlantic storm track (Hoskins and Hodges, 2005; Gramscianinov et al., 2019), whereas EOF1(-) systems remain more confined near the South American

775 coast (Figure 10a), particularly within the ARG and SE-BR regions (Figure 11d), and
 776 exhibit lower translational speeds (Figure S1).

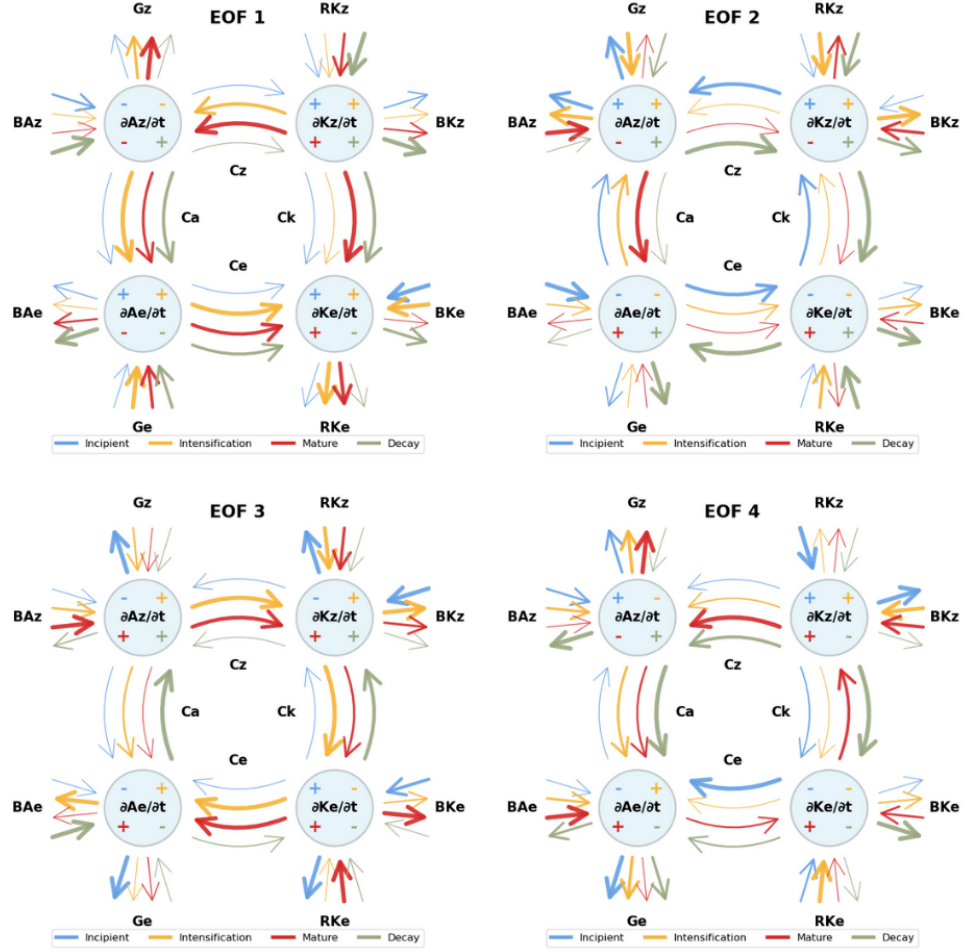


Fig. 16 Schematic representation of the mean Lorenz Energy Cycle (LEC) for cyclones associated with the four leading positive Empirical Orthogonal Functions (EOF1(+) to EOF4(+)). Arrows represent energy fluxes between reservoirs, with colors indicating life cycle phases. The thickness of each arrow and the size of the balance terms ($\partial/\partial t$) symbols are proportional to the absolute magnitude of each term.

777 EOF2–EOF4 describe less frequent cyclone types. EOF2(+), EOF3(–), and
 778 EOF4(–) are associated with relatively stronger systems, while their counterparts cor-
 779 respond to weaker ones (Figure S1b). EOF2(+) is notable for lacking enhancements
 780 in baroclinic/barotropic conversions or K_E imports during early phases. EOF3(–) and
 781 EOF4(–), in contrast, display a pronounced intensification of the $G_E \rightarrow A_E \rightarrow K_E$

chain, especially during the incipient and intensification phases. These patterns indicate that despite a shared dominant energy structure, South Atlantic cyclones express diverse configurations in their development. This variability is further reflected in differences in genesis region and seasonality (Figure 11), with EOF1(+) and EOF3(-) dominating among the most intense systems (Supplementary Figure S3).

For **corrected** intense cyclones (above the 90th percentile in central vorticity), genesis is most frequent in **ARG** (53.3%), followed by **LA-PLATA** and **during JJA**—consistent with earlier studies identifying this region as favorable for strong development. These systems are primarily linked to moist baroclinic instability (Figure 12), particularly under the influence of a strong upper-level jet in austral winter. Meanwhile, (28.7%) and **SE-BR** systems are less frequent, especially during **DJF**, when baroclinicity in that region is weaker (18.0%). **JJA** is the most frequent season (27.4%), closely followed by **SON** (26.1%). These systems retain a strong moist-baroclinic pathway, while the corrected cluster memberships and pathway proportions require the replacement cluster figures for final interpretation.

~~Among the four~~The corrected reconstruction contains five, not four, LEC-based clusters, ~~Cluster 4~~. In the canonical matched numbering, Group 1 contains the largest number of ~~systems but the lowest~~ corrected systems (647) and the lowest mean maximum central vorticity, ~~making it the weakest group~~. Importantly, all clusters show enhanced LEC terms relative to the overall mean, reinforcing the connection between stronger energy fluxes and cyclone intensity. These findings support the robustness of the Semi-Lagrangian framework and highlight the existence of distinct energetic regimes. Even within the subset of $(9.84 \times 10^{-5} \text{ s}^{-1})$. The five-group solution continues to demonstrate diversity among intense cyclones, ~~considerable diversity in energy cycle structures persists~~, indicating that multiple pathways can lead to strong cyclogenesis in the South Atlantic but changes in membership and in the affected conversion terms preclude carrying over the published cluster-specific physical narratives unchanged.

4.3 Mechanisms of Energy Transfer

~~Throughout this study, barotropic conversions~~The corrected results identify baroclinic conversion (C_A), rather than barotropic conversion (C_K) ~~emerged as a key~~, as the dominant conversion mechanism for cyclone development in the South ~~American region~~. While extratropical cyclones are traditionally associated with baroclinic instability, our results show that moist baroclinic processes dominate during the intensification phase, whereas barotropic conversions peak during the mature stage (Figure 4). ~~Notably, Atlantic~~. In the correction-only paired control, median C_A increases from 0.50 to 1.54 W m^{-2} , whereas median C_K ~~magnitudes are nearly~~ shifts from -1.91 to -0.03 W m^{-2} . The proportion of cyclone phases in which an eddy-feeding C_K exceeds C_A falls from 64.7% to 29.3%, and to 19.2% during intensification. Barotropic conversion remains an eddy-energy source in a substantial subset of cases, particularly later in the lifecycle, but it is neither ubiquitous nor typically two to three times larger than ~~those of C_A , suggesting a stronger contribution from zonal kinetic energy to eddy development than from meridional temperature gradients~~baroclinic conversion.

Despite LEC-based studies dating back decades, few have examined South Atlantic cyclones, limiting direct comparisons. Case studies on tropical systems emphasize the

$G_E \rightarrow A_E \rightarrow K_E$ chain and barotropic conversions (Brennan and Vincent, 1980; Veiga et al., 2008), while subtropical systems show mixed roles for both baroclinic and barotropic conversions (Michaelides, 1987; Dias Pinto et al., 2013; Pezza et al., 2014; Cavicchia et al., 2018). For extratropical systems, energy pathways vary: some are baroclinically dominated (Dias Pinto and Rocha, 2011; Black and Pezza, 2013), others barotropically (Michaelides, 1992; Dias Pinto and Rocha, 2011), and some combine both with varying degrees of K_E import (Wahab et al., 2002; Bulic, 2006; Pezza et al., 2010). Thus, although the relevance of barotropic instability in extratropical cyclones is not novel, this work provides the first large-sample LEC analysis for the South Atlantic, offering a consolidated view of this behavior. These findings are consistent with recent studies that link storm track intensification in the Southern Hemisphere to enhanced barotropic growth, associate storm growth time with barotropic conversion, and identify barotropic conversions from K_Z to K_E within storm track regions. The corrected climatology supports a dominant baroclinic pathway with a variable, secondary barotropic contribution; it does not support a consolidated picture of barotropic dominance.

It is also important to emphasize that, while previous studies investigated the LEC using an Eulerian framework, this study employed a Semi-Lagrangian framework. This approach has the advantage of isolating the energetics strictly related to the target system. However, the extent to which this methodology differs from the Eulerian framework in representing the magnitude and direction of energy fluxes has not been extensively assessed, apart from one study by Michaelides et al. (1999). Also, the extent to which our findings generalize to other regions remains uncertain. Given the stronger jets and meridional wind gradients in the Southern Hemisphere, barotropic conversions may be. The corrected results do not provide evidence that barotropic conversion is more pronounced in the South Atlantic than in the North Atlantic—a hypothesis requiring further study.

Another distinction is found in the decay phase. While most previous studies report $K_E \rightarrow K_Z$ conversions during cyclone decay, this pattern was not observed here. In our analysis, the decay phase, the longest for these systems, is averaged over multiple timesteps and features. The corrected C_K values that gradually shift from strongly negative in the mature phase to near neutral at the end of the lifecycle (Figures 4C and S4). This suggests that barotropic conversion continues to sustain the system during decay, until dissipation dominates. Prior use of large Eulerian domains may have underestimated C_K , as seen in, where C_K remains positive across all phases. evolution also changes the comparison with earlier decay-phase studies. In the paired control, phase medians are positive during incipience and intensification (0.53 and 0.10 W m^{-2}) and modestly negative during maturity and decay (-0.65 and -0.75 W m^{-2}). The published progression from strongly eddy-feeding conversion in maturity toward neutrality during decay is therefore not reproduced. Barotropic conversion can feed eddies during later phases, but the corrected statistics do not support the claim that it continuously sustains most systems through decay.

Some limitations of our approach must be acknowledged. Although the Semi-Lagrangian framework follows the cyclone’s trajectory, it does not account for regional variations in background APE and kinetic energy. This may influence A_Z , K_Z , and

their boundary terms as the system moves through varying energy environments. Furthermore, the LEC formulation relies on a global reference state for APE. In contrast, the local APE framework proposed by Novak and Tailleux (2018) defines APE relative to each air parcel’s individual reference state. While this approach can yield additional insights—such as the potential for cyclones to locally generate APE (Federer et al., 2024), the overall global energy fluxes remain consistent between the global and local formulations (Liu et al., 2024). Moreover, the modest BA_Z values observed here (Figure 4) may be partially explained by APE transport from the polar upper troposphere, as shown for the North Atlantic storm tracks by Federer et al. (2025).

5 Summary and Conclusions

In this study, we examined the Lorenz Energy Cycle (LEC) for a validated corrected population of 3,820 of the 6,789 archived cyclonic systems originating in the Southwestern Atlantic, ~~presenting what is, to the best of our knowledge, the first comprehensive climatology of cyclones’ energy cycles in this region.~~ Comparisons with the full published population therefore combine the toolkit correction and a population change. A notable exception to this large-sample climatological approach is the work by Black and Pezza (2013), which focused solely on explosive cyclones and used large fixed domains, contrasting with the Semi-Lagrangian approach adopted in this study. While studies such as Smith (1980) provide valuable reviews of the energetics of extratropical cyclones, focusing on broad averages and generalized mechanisms, and case studies have explored the dynamics of specific systems (Pezza et al., 2014; Dias Pinto and Rocha, 2011; Cavicchia et al., 2018, e.g.), our study ~~adopts a climatological approach. Specifically, we track tracks~~ the energy cycles of individual cyclones across distinct life cycle phases, ~~offering a more detailed investigation of the unique physical mechanisms that drive energy conversions in this region. Furthermore, the use of the Cyclophaser program to dissect the life cycles of cyclones has enabled a novel investigation of the energy cycle across different developmental stages, providing deeper insights into the variability and dynamism of these systems.~~

The ~~results presented here demonstrate that, corrected results demonstrate a clear baroclinic energy pathway for most cyclonic systems in the South Atlantic, a clear pattern of energy flow is evident. This pattern is characterized by both barotropic and baroclinic conversions providing energy for eddy development and is preserved throughout the cyclone life cycle phases, despite variations in magnitude. Barotropic conversions tend to be 2 to 3 times larger in magnitude than baroclinic conversions, with the former peaking during the mature phase and the latter during the intensification phase. The.~~ Baroclinic conversion peaks during intensification and is the dominant conversion throughout the mean lifecycle. Barotropic conversion is close to neutral in the pooled median, changes sign across phases and cases, and exceeds baroclinic conversion in only 29.3% of paired cyclone phases. It remains a relevant pathway for a subset of systems, but the published claim that it is typically two to three times larger than baroclinic conversion does not survive the correction. The unchanged generation of eddy potential energy ~~and the imports of eddy kinetic~~

energy play secondary roles. The former peaks during the intensification phase and remains positive through the mature and decay phases, while the latter occurs peaks during intensification, while eddy kinetic energy imports occur primarily during the incipient (cyclogenesis) and intensification phases.

Despite the high variability among the systems, the EOF analysis indicates that, in most cases, the main behavior is preserved. This variability is instead expressed as a strengthening and/or weakening of specific energy pathways, indicating that cyclonic development in the South Atlantic region can be attributed to distinct dynamical processes, as supported by the literature. Consequently, it is possible to hypothesize that these systems can be grouped into distinct types based on their characteristics and energy cycles. leading corrected EOF remains recognizable, while higher modes can exchange rank and must be compared by matched loading pattern. This heterogeneity is also evident among the most intense systems. For these systems, distinct groups exhibit varying relative contributions of energy from barotropic and baroclinic chains, as well as differences in eddy generation of APE and imports of eddy kinetic energy. Although most of these intense, for which the corrected reconstruction retains five distinct groups. Across that corrected subset, most systems originate in the ARG, although LA-PLATA region, the variability in the relative proportions of genesis regions among the different clusters is reflected in their energy cycles and mean characteristics is overrepresented relative to its share of the full published dataset. Final cluster-specific physical interpretations require the corrected-only replacement figures and author review.

The goal of this study was to investigate the energy cycle of cyclones in the South Atlantic region using a Semi-Lagrangian approach. As a pioneering study of this kind, several questions have emerged throughout its development. For instance, previous Previous studies for this region have assessed the distinct synoptic and dynamical conditions related to cyclogenesis in the three primary genesis regions, as well as the seasonal variability in their development (Gramscianinov et al., 2019; Crespo et al., 2021). A distinction in the energy cycles among these genesis regions, along with their seasonal variability, would therefore be valuable. Additionally, given the prominence of barotropic conversions as an energy source for cyclone development, the question arises as to whether barotropic instability is truly occurring remains valuable. The corrected results motivate investigation of when and where the weaker, case-dependent barotropic pathway feeds eddy kinetic energy, rather than assuming that it is a ubiquitous dominant source near the cyclone center. proposed a framework for classifying cyclones into distinct groups based on their energy cycles. In an upcoming study, this method will be applied to address these questions and further investigate these topics.

Appendix

Mathematical expressions used for the calculation of the Lorenz Energy Cycle (LEC), adopting the notation from Michaelides (1987).

Firstly, we define the zonal mean of a variable X , between longitudes λ_1 and λ_2 :

$$[X]_{\lambda} = \frac{1}{\lambda_2 - \lambda_1} \int_{\lambda_2}^{\lambda_1} X d\lambda \quad (13)$$

955 The eddy component of this variable is its deviation from the zonal mean:

$$(X)_{\lambda} = X - [X]_{\lambda} \quad (14)$$

956 The domain mean of the variable X , defined over the computational domain
957 bounded by longitudes λ_1 and λ_2 , and latitudes ϕ_1 and ϕ_2 , is given by:

$$[X]_{\lambda\phi} = \left(\frac{1}{\lambda_2 - \lambda_1} \right) \left(\frac{1}{\sin \phi_2 - \sin \phi_1} \right) \int_{\lambda_2}^{\lambda_1} X \cos \phi d\lambda d\phi \quad (15)$$

958 Similarly, we define the deviation of the zonal mean from the domain mean:

$$([X]_{\lambda})_{\phi} = [X]_{\lambda} - [X]_{\lambda\phi} \quad (16)$$

959 From the definitions above, the four energy components used in the LEC
960 computation are defined as follows:

$$A_Z = \int_{p_t}^{p_b} \frac{[(T)_{\lambda}]_{\phi}^2}{2[\sigma]_{\lambda\phi}} dp \quad (17)$$

$$A_E = \int_{p_t}^{p_b} \frac{[(T)_{\lambda}]_{\lambda\phi}^2}{2[\sigma]_{\lambda\phi}} dp \quad (18)$$

$$K_Z = \int_{p_t}^{p_b} \frac{[(u)_{\lambda}]^2 + [(v)_{\lambda}]^2}{2g} dp \quad (19)$$

$$K_E = \int_{p_t}^{p_b} \frac{[(u)_{\lambda}]^2 + [(v)_{\lambda}]^2}{2g} dp \quad (20)$$

961 where p is the atmospheric pressure, with subscripts b and t denoting the lower
962 (base) and upper (top) pressure boundaries of the atmosphere, respectively. T repre-
963 sents temperature, g is the acceleration due to gravity, and u and v are the zonal and
964 meridional wind components, respectively. The static stability parameter σ is defined
965 as:

$$\sigma = \left[\frac{gT}{c_p} - \frac{pg}{R} \frac{\partial T}{\partial p} \right]_{\lambda\phi} \quad (21)$$

966 where c_p is the specific heat at constant pressure, and R is the ideal gas constant
967 for dry air.

968 The four conversion terms are defined as follows, integrating over the atmospheric
969 column from the base (p_b) to the top (p_t) pressures:

$$C_Z = \int_{p_t}^{p_b} -[([T]_{\lambda})_{\phi}([\omega]_{\lambda})_{\phi}]_{\lambda\phi} \frac{R}{gp} dp \quad (22)$$

$$C_E = \int_{p_t}^{p_b} -[(T)_\lambda(\omega)_\lambda]_{\lambda\phi} \frac{R}{g p} dp \quad (23)$$

$$C_A = \int_{p_t}^{p_b} - \left(\frac{1}{2a\sigma} \frac{1}{a\sigma} \left[(v)_\lambda(T)_\lambda \frac{\partial([T]_\lambda)_\phi}{\partial\phi} \right]_{\lambda\phi} + \frac{1}{\sigma} \left[(\omega)_\lambda(T)_\lambda \frac{\partial([T]_\lambda)_\phi}{\partial p} \right]_{\lambda\phi} \right) dp \quad (24)$$

$$C_K = \int_{p_t}^{p_b} \frac{1}{g} \left(\left[\frac{\cos\phi}{a} (u)_\lambda(v)_\lambda \frac{\partial}{\partial\phi} \left(\frac{[u]_\lambda}{\cos\phi} \right) \right]_{\lambda\phi} + \left[\frac{(v)_\lambda^2}{a} \frac{\partial[v]_\lambda}{\partial\phi} \right]_{\lambda\phi} + \left[\frac{\tan\phi}{a} (u)_\lambda^2 [v]_\lambda \right]_{\lambda\phi} + \left[(\omega)_\lambda(u)_\lambda \frac{\partial[u]_\lambda}{\partial p} \right]_{\lambda\phi} + \left[(\omega)_\lambda(v)_\lambda \frac{\partial[v]_\lambda}{\partial p} \right]_{\lambda\phi} \right) dp \quad (25)$$

where a is the Earth's radius and ω is the vertical velocity in isobaric coordinates.
The APE generation and K dissipation terms are defined as:

$$G_Z = \int_{p_t}^{p_b} \frac{[(q)_\lambda]_\phi ([T]_\lambda)_\phi]_{\lambda\phi}}{c_p [\sigma]_{\lambda\phi}} dp \quad (26)$$

$$G_E = \int_{p_t}^{p_b} \frac{[(q)_\lambda(T)_\lambda]_{\lambda\phi}}{c_p [\sigma]_{\lambda\phi}} dp \quad (27)$$

$$D_Z = - \int_{p_t}^{p_b} \frac{1}{g} [[u]_\lambda [F_\lambda]_\lambda + [v]_\lambda [F_\phi]_\lambda]_{\lambda\phi} dp \quad (28)$$

$$D_E = - \int_{p_t}^{p_b} \frac{1}{g} [(u)_\lambda (F_\lambda)_\lambda + (v)_\lambda (F_\phi)_\lambda]_{\lambda\phi} dp \quad (29)$$

Here, F_λ and F_ϕ represent the zonal and meridional frictional components, respectively, and q is the diabatic heating term, computed as a residual from the thermodynamic equation:

$$\frac{q}{c_p} = \frac{\partial T}{\partial t} + \vec{V}_H \cdot \vec{\nabla}_p T - S_p \omega \quad (30)$$

where $-\vec{V}_H - \vec{\nabla}_p T - \vec{V}_H \cdot \vec{\nabla}_p T$ represents the horizontal advection of temperature and S_p approximates the static stability, given by:

$$S_p \equiv - \frac{T}{\theta} \frac{\partial \theta}{\partial p} \quad (31)$$

where θ is the potential temperature.
The boundary terms are given by:

$$\begin{aligned} \text{BAZ} = & c_1 \int_{p_1}^{p_2} \int_{\varphi_1}^{\varphi_2} \frac{1}{2[\sigma]_{\lambda\varphi}} \left(2([T]_\lambda)_\varphi (T)_\lambda u + ([T]_{\lambda\varphi})_\varphi^2 u \right)_{\lambda_1}^{\lambda_2} \\ & \times d\varphi dp + c_2 \int_{p_1}^{p_2} \frac{1}{2[\sigma]_{\lambda\varphi}} \left(2[(v)_\lambda(T)_\lambda]_\lambda ([T]_\lambda)_\varphi \cos\varphi + ([T]_\lambda)_\varphi^2 [v]_\lambda \cos\varphi \right)_{\varphi_1}^{\varphi_2} dp \end{aligned}$$

$$- \frac{1}{2[\sigma]_{\lambda\varphi}} \left([2(\omega)_{\lambda}(T)_{\lambda}]_{\lambda} ([T]_{\lambda})_{\varphi} + [\omega]_{\lambda} ([T]_{\lambda})_{\varphi}^2 \right)_{\lambda\varphi}^{p_2} \quad (32)$$

$$\begin{aligned} \text{BAE} &= c_1 \int_{p_1}^{p_2} \int_{\varphi_1}^{\varphi_2} \frac{1}{2[\sigma]_{\lambda\varphi}} [u(T)_{\lambda}^2]_{\lambda_1}^{\lambda_2} d\varphi dp \\ &+ c_2 \int_{p_1}^{p_2} \frac{1}{2[\sigma]_{\lambda\varphi}} \left([(T)_{\lambda}^2 v]_{\lambda} \cos \varphi \right)_{\varphi_1}^{\varphi_2} dp \\ &- \left(\frac{[\omega(T)_{\lambda}^2]_{\lambda\varphi}}{2[\sigma]_{\lambda\varphi}} \right)_{p_1}^{p_2} \end{aligned} \quad (33)$$

$$\begin{aligned} \text{BKZ} &= c_1 \int_{p_1}^{p_2} \int_{\varphi_1}^{\varphi_2} \frac{1}{2g} \left(u [u^2 + v^2 - (u)_{\lambda}^2 - (v)_{\lambda}^2] \right)_{\lambda_1}^{\lambda_2} \\ &\times d\varphi dp + c_2 \int_{p_1}^{p_2} \frac{1}{2g} \left([v \cos \varphi [u^2 + v^2 - (u)_{\lambda}^2 - (v)_{\lambda}^2]]_{\varphi_1}^{\varphi_2} \right) dp \\ &- \left(\frac{1}{2g} [\omega [u^2 + v^2 - (u)_{\lambda}^2 - (v)_{\lambda}^2]]_{\lambda\varphi} \right)_{p_1}^{p_2} \end{aligned} \quad (34)$$

$$\begin{aligned} \text{BKE} &= c_1 \int_{p_1}^{p_2} \int_{\varphi_1}^{\varphi_2} \frac{1}{2g} \left(u [(u)_{\lambda}^2 + (v)_{\lambda}^2] \right)_{\lambda_1}^{\lambda_2} d\varphi dp \\ &+ c_2 \int_{p_1}^{p_2} \frac{1}{2g} \left([v \cos \varphi [(u)_{\lambda}^2 + (v)_{\lambda}^2]]_{\lambda} \right)_{\varphi_1}^{\varphi_2} dp \\ &- \left(\frac{1}{2g} [\omega [(u)_{\lambda}^2 + (v)_{\lambda}^2]]_{\lambda\varphi} \right)_{p_1}^{p_2} \end{aligned} \quad (35)$$

979 where $c_1 = -[a(\lambda_2 - \lambda_1)(\sin \varphi_2 - \sin \varphi_1)]^{-1}$, $c_2 = -[a - x(\sin \varphi_2 - \sin \varphi_1)]^{-1} c_1 =$
 980 $-[a(\lambda_2 - \lambda_1)(\sin \varphi_2 - \sin \varphi_1)]^{-1}$ and $c_2 = -[a(\sin \varphi_2 - \sin \varphi_1)]^{-1}$.
 981 Lastly, the terms $B\Phi Z$ and $B\Phi E$ are given by:

$$\begin{aligned} B\Phi Z &= c_1 \int \frac{p_2 p_b}{p_1 p_t} \int \frac{\varphi_2 \varphi_n}{\varphi_1 \varphi_s} \frac{1}{g} ([\mathbf{v}u]_{\lambda} \Delta_{EW} \Phi_{\lambda'} + \Phi^* \Delta_{EW} u') \frac{\lambda_2}{\varphi \lambda_1} d\varphi dp \\ &+ c_2 \int \frac{p_2 p_b}{p_1 p_t} \frac{1}{g} (\cos \varphi [v]_{\lambda} \Phi_{\lambda}^*) \frac{\varphi_2 \varphi_n}{\varphi \varphi_1 \varphi_s} dp \\ &- \frac{1}{g} \left([\omega \lambda \varphi^* \Phi_{\lambda \varphi}^*]_{\lambda \varphi} \right) \frac{p_2 p_b}{p_1 p_t} \end{aligned} \quad (36)$$

$$\begin{aligned} B\Phi E &= c_1 \int_{p_1}^{p_2} \int_{\varphi_1}^{\varphi_2} \frac{1}{g} ((u)_{\lambda}(\Phi)_{\lambda})_{\lambda_1}^{\lambda_2} d\varphi dp \\ &+ c_2 \int_{p_1}^{p_2} \frac{1}{g} \left([(v)_{\lambda}(\Phi)_{\lambda}]_{\lambda} \cos \varphi \right)_{\varphi_1}^{\varphi_2} dp \\ &- \frac{1}{g} \left([(\omega)_{\lambda}(\Phi)_{\lambda}]_{\lambda\varphi} \right)_{p_1}^{p_2} \end{aligned} \quad (37)$$

982 Here, $\Phi^* = [\Phi]_\lambda - [\Phi]_{\lambda\varphi}$, $\omega^* = [\omega]_\lambda - [\omega]_{\lambda\varphi}$, $\Phi' = \Phi - [\Phi]_\lambda$, $u' = u - [u]_\lambda$, and
983 $\Delta_{EW}X = X|_{\lambda_2} - X|_{\lambda_1}$.

984 Declarations

985 Funding

986 This study was partly financed by the Coordenação de Aperfeiçoamento de Pessoal
987 de Nível Superior – Brasil (CAPES) under Finance Code 001.

988 Competing interests

989 The authors have no relevant financial or non-financial interests to disclose.

990 Data Availability

991 The cyclone tracks used in this study were obtained from the *Atlantic Extratropical Cyclone Tracks Database*, available at <https://doi.org/10.17632/kwcvfr52hp>.
992 4. ~~All datasets, including the computed Lorenz Energy Cycle results for each~~
993 ~~eyclone, along with the scripts used to generate the figures and analyses presented~~
994 ~~in this study, are publicly available at the following GitHub repository: The~~
995 ~~archived article inputs and scripts are available at~~ https://github.com/daniloceano/energetic_patterns_cyclones_south_atlantic. The corrected calculation, validation products, statistics, figures, and provenance used for this correction are maintained
996 at <https://github.com/daniloceano/lec-climatology-rerun>. The source code of the
997 Cyclophaser package, ~~used for detecting cyclone life cycle phases,~~ is available on PyPI
998 at <https://pypi.org/project/cyclophaser/>. ~~The LorenzCycleToolkit package, used~~
999 ~~for computing the Lorenz Energy Cycle components, is also available on PyPI at,~~ and
1000 the corrected LorenzCycleToolkit is available at <https://github.com/daniloceano/LorenzCycleToolkit>.
1001
1002
1003
1004

1005 Author contributions

1006 Conceptualization: Danilo Couto de Souza, Pedro Leite da Silva Dias, Ricardo de
1007 Camargo, Carolina Barnez Gramscianinov; Methodology: Danilo Couto de Souza,
1008 Pedro Leite da Silva Dias, Ricardo de Camargo; Formal analysis and investigation:
1009 Danilo Couto de Souza, Pedro Leite da Silva Dias; Writing - original draft preparation:
1010 Danilo Couto de Souza; Writing - review and editing: Danilo Couto de Souza, Pedro
1011 Leite da Silva Dias, Ricardo de Camargo, Carolina Barnez Gramscianinov; Funding
1012 acquisition: Pedro Leite da Silva Dias, Ricardo de Camargo; Resources: Pedro Leite
1013 da Silva Dias, Ricardo de Camargo; Supervision: Pedro Leite da Silva Dias, Ricardo
1014 de Camargo.

1015 References

1016 Andrade, H.N., A.B. Nunes, M.S. Teixeira, M.F.L. de Quadro, V.D. de Avila, F.S.C.
1017 de Oliveira, and R.d.C.M. Alves. 2025. Composite analysis of explosive cyclones in

- the southern atlantic ocean. *International Journal of Climatology* 45(4): e8737 .
- Andrade, H.N., M.F.L.D. Quadro, A.B. Nunes, F.S.D. Oliveira, V.D.D. Avila, M.S. Teixeira, and R.D.C.M. Alves. 2024. Explosive cyclones occurred between 2010 and 2020 in the south atlantic under the perspective of two detection schemes. *Anais da Academia Brasileira de Ciências* 96(suppl 1): e20231051 .
- Bengfort, B. and R. Bilbro. 2019. Yellowbrick: Visualizing the scikit-learn model selection process. *Journal of Open Source Software* 4(35): 1075 .
- Black, M.T. and A.B. Pezza. 2013. A universal, broad-environment energy conversion signature of explosive cyclones. *Geophysical Research Letters* 40(2): 452–457 .
- Brennan, F.E. and D.G. Vincent. 1980. Zonal and eddy components of the synoptic-scale energy budget during intensification of hurricane carmen (1974). *Monthly Weather Review* 108(7): 954–965. [https://doi.org/10.1175/1520-0493\(1980\)108<0954:ZAECOT>2.0.CO;2](https://doi.org/10.1175/1520-0493(1980)108<0954:ZAECOT>2.0.CO;2) .
- Bulic, I. 2006. Limited area energy budget during a life cycle of genoa cyclone (18-21 november 1999). *NUOVO CIMENTO-SOCIETA ITALIANA DI FISICA SEZIONE C* 29(2): 167 .
- Cardoso, A.A., R.P. da Rocha, and N.M. Crespo. 2022. Synoptic climatology of subtropical cyclone impacts on near-surface winds over the south atlantic basin. *Earth and Space Science* 9(11): e2022EA002482 .
- Cavicchia, L., A. Dowdy, and K. Walsh. 2018. Energetics and dynamics of subtropical australian east coast cyclones: Two contrasting cases. *Monthly Weather Review* 146(5): 1511–1525 .
- Crespo, N.M., R.P. da Rocha, M. Sprenger, and H. Wernli. 2021. A potential vorticity perspective on cyclogenesis over centre-eastern south america. *International Journal of Climatology* 41(1): 663–678 .
- da Silva, M.B.L., D.C. de Souza, F.T.C. Barreto, R. Tecchio, R. de Freitas Pimentel dos Anjos, C.B. Gramscianinov, and R. De Camargo. 2025. Classification of synoptic weather patterns associated with extreme wave events in different regions of western south atlantic. *Natural Hazards*: 1–25 .
- de Souza, D. and R.R. da Silva. 2021. Ocean-land atmosphere model (olam) performance for major extreme meteorological events near the coastal region of southern brazil. *Climate Research* 84: 1–21 .
- de Souza, D.C., N.M. Crespo, D.V. da Silva, L.M. Harada, R.M.P. de Godoy, L.M. Domingues, R. Luiz, C.A. Bortolozo, D. Metodiev, M.R.M. de Andrade, et al. 2024. Extreme rainfall and landslides as a response to human-induced climate change: a case study at baixada santista, brazil, 2020. *Natural Hazards*: 1–26 .

- de Souza, D.C., P.L.S. da Dias, M.B.L. da Silva, and R. de Camargo. 2024. Lorenz-cycletoolkit: A comprehensive python tool for analyzing atmospheric energy cycles. *Journal of Open Source Software* 9(101): 7139 .
- de Souza, D.C., P.L.S. da Dias, C.B. Gramscianinov, and R. de Camargo. 2025. Cyclophaser: A python package for detecting extratropical cyclone life cycles. *Journal of Open Source Software* 10(108): 7363 .
- de Souza, D.C., P.L. da Silva Dias, C.B. Gramscianinov, M.B.L. da Silva, and R. de Camargo. 2024. New perspectives on south atlantic storm track through an automatic method for detecting extratropical cyclones' lifecycle. *International Journal of Climatology* 44(10): 3568–3588 .
- de Souza, D.C., R. Ramos da Silva, P. Gomes da Silva, A.F.H. Fetter Filho, F.J. Mendez, and D. Werth. 2022. A hybrid regional climate downscaling for the southern brazil coastal region. *International Journal of Climatology* 42(13): 6753–6770 .
- de Souza, D.C., V.A. Ranieri, P.L. da Silva Dias, A.R. Flores, and R. de Camargo. 2025. Revisiting the rare transition of a south atlantic cyclone to tropical storm akar : Energy cycle and stratosphere-troposphere interaction. manuscript submitted to Climate Dynamics. Preprint available at <https://scity.org/articles/activity/10.21203/rs.3.rs-6536558/v1>.
- Dias Pinto, J.R., M.S. Reboita, and R.P. da Rocha. 2013. Synoptic and dynamical analysis of subtropical cyclone anita (2010) and its potential for tropical transition over the south atlantic ocean. *Journal of Geophysical Research: Atmospheres* 118(19): 10–870. <https://doi.org/10.1002/jgrd.50830> .
- Dias Pinto, J.R. and R.P. Rocha. 2011. The energy cycle and structural evolution of cyclones over southeastern south america in three case studies. *Journal of Geophysical Research: Atmospheres* 116(D14) .
- Dutton, J.A. and D.R. Johnson. 1967. The theory of available potential energy and a variational approach to atmospheric energetics. *Advances in geophysics* 12: 333–436 .
- Federer, M., L. Papritz, M. Sprenger, and C.M. Grams. 2025. Synoptic perspective on the conversion and maintenance of local available potential energy in extratropical cyclones. *Weather and Climate Dynamics* 6(1): 211–230 .
- Federer, M., L. Papritz, M. Sprenger, C.M. Grams, and M. Wenta. 2024. On the local available potential energy perspective of baroclinic wave development. *Journal of the Atmospheric Sciences* 81(5): 871–886 .
- Fukuoka, A. 1951. A study of 10-day forecast (a synthetic report). *The Geophysical Magazine* XXII: 177–218 .

- 1091 Govekar, P.D., C. Jakob, M.J. Reeder, and J. Haynes. 2011. The three-dimensional dis-
1092 tribution of clouds around southern hemisphere extratropical cyclones. *Geophysical*
1093 *Research Letters* 38(21). <https://doi.org/10.1029/2011GL049091> .
- 1094 Gramscianinov, C., R. Campos, R. De Camargo, K. Hodges, C.G. Soares, and
1095 P. da Silva Dias. 2020. Analysis of atlantic extratropical storm tracks characteristics
1096 in 41 years of era5 and cfsr/cfsv2 databases. *Ocean Engineering* 216: 108–111 .
- 1097 Gramscianinov, C., K. Hodges, and R.d. Camargo. 2019. The properties and genesis
1098 environments of south atlantic cyclones. *Climate Dynamics* 53: 4115–4140 .
- 1099 Gramscianinov, C.B., R.M. Campos, R. de Camargo, K.I. Hodges, C. Guedes Soares,
1100 and P.L. da Silva Dias. 2020. Atlantic extratropical cyclone tracks in 41 years of
1101 ERA5 and CFSR/CFSv2 databases. *Mendeley Data V4*. [https://doi.org/10.17632/](https://doi.org/10.17632/kwcvfr52hp.4)
1102 [kwcvfr52hp.4](https://doi.org/10.17632/kwcvfr52hp.4) .
- 1103 Gramscianinov, C.B., R. de Camargo, R.M. Campos, C. Guedes Soares, and P.L.
1104 da Silva Dias. 2023. Impact of extratropical cyclone intensity and speed on the
1105 extreme wave trends in the atlantic ocean. *Climate dynamics* 60(5-6): 1447–1466 .
- 1106 Gutierrez, E.R., P.S. Dias, J. Veiga, R. Camayo, and A. Dos Santos. 2009. Multivariate
1107 analysis of the energy cycle of the south american rainy season. *International*
1108 *Journal of Climatology* 29(15): 2256–2269 .
- 1109 Hao, J. and T.K. Ho. 2019. Machine learning made easy: a review of scikit-learn
1110 package in python programming language. *Journal of Educational and Behavioral*
1111 *Statistics* 44(3): 348–361 .
- 1112 Hartigan, J.A. and M.A. Wong. 1979. Algorithm as 136: A k-means clustering algo-
1113 rithm. *Journal of the Royal Statistical Society. Series C (Applied Statistics)* 28(1):
1114 100–108 .
- 1115 Hernández-Deckers, D. and J.S. von Storch. 2010. Energetics responses to increases
1116 in greenhouse gas concentration. *Journal of Climate* 23(14): 3874–3887 .
- 1117 Hersbach, H., B. Bell, P. Berrisford, S. Hirahara, A. Horányi, J. Muñoz-Sabater,
1118 J. Nicolas, C. Peubey, R. Radu, D. Schepers, et al. 2020. The era5 global reanalysis.
1119 *Quarterly Journal of the Royal Meteorological Society* 146(730): 1999–2049 .
- 1120 Hodges, K. 1995. Feature tracking on the unit sphere. *Monthly Weather*
1121 *Review* 123(12): 3458–3465 .
- 1122 Hodges, K.I. 1994. A general method for tracking analysis and its application to
1123 meteorological data. *Monthly Weather Review* 122(11): 2573–2586 .
- 1124 Hoskins, B.J. and K.I. Hodges. 2002. New perspectives on the northern hemisphere
1125 winter storm tracks. *Journal of the Atmospheric Sciences* 59(6): 1041–1061. [https://doi.org/10.1175/1520-6932\(2002\)059<1041:NPO](https://doi.org/10.1175/1520-6932(2002)059<1041:NPO)

1126 //doi.org/10.1175/1520-0469(2002)059<1041:NPOTNH>2.0.CO;2 .

1127 Hoskins, B.J. and K.I. Hodges. 2005. A new perspective on southern hemisphere
1128 storm tracks. *Journal of Climate* 18(20): 4108–4129 .

1129 Hu, Q., Y. Tawaye, and S. Feng. 2004. Variations of the northern hemisphere
1130 atmospheric energetics: 1948–2000. *Journal of Climate* 17(10): 1975–1986 .

1131 Keshtgar, B., A. Voigt, C. Hoose, M. Riemer, and B. Mayer. 2023. Cloud-
1132 radiative impact on the dynamics and predictability of an idealized extratropical
1133 cyclone. *Weather and Climate Dynamics* 4(1): 115–132. [https://doi.org/10.5194/](https://doi.org/10.5194/wcd-4-115-2023)
1134 wcd-4-115-2023 .

1135 Kirshbaum, D., T. Merlis, J. Gyakum, and R. McTaggart-Cowan. 2018. Sensitivity
1136 of idealized moist baroclinic waves to environmental temperature and moisture
1137 content. *Journal of the Atmospheric Sciences* 75(1): 337–360 .

1138 Lau, N.C. and M.W. Crane. 1997. Comparing satellite and surface observa-
1139 tions of cloud patterns in synoptic-scale circulation systems. *Monthly weather*
1140 *review* 125(12): 3172–3189. [https://doi.org/10.1175/1520-0493\(1997\)125\(3172:](https://doi.org/10.1175/1520-0493(1997)125(3172:CSASOO)2.0.CO;2)
1141 CSASOO)2.0.CO;2 .

1142 Leal, K.B., L.E.d.S. Robaina, T.S. Körting, J.L. Nicolodi, J.D. da Costa, and V.G.
1143 Souza. 2023. Identification of coastal natural disasters using official databases to
1144 provide support for the coastal management: the case of santa catarina, brazil.
1145 *Natural Hazards*: 1–18 .

1146 Li, L., A.P. Ingersoll, X. Jiang, D. Feldman, and Y.L. Yung. 2007. Lorenz energy
1147 cycle of the global atmosphere based on reanalysis datasets. *Geophysical Research*
1148 *Letters* 34(16) .

1149 Liu, Z., C.L. Franzke, L. Novak, R. Tailleux, and V. Lembo. 2024. A systematic
1150 local view of the long-term changes in the atmospheric energy cycle. *Journal of*
1151 *Climate* 37(23): 6413–6434 .

1152 Lorenz, E. 1967. The nature and theory of the general circulation of the atmosphere.
1153 *World meteorological organization* 161 .

1154 Lorenz, E.N. 1955. Available potential energy and the maintenance of the general
1155 circulation. *Tellus* 7(2): 157–167 .

1156 Lorenz, E.N. 1956. *Empirical orthogonal functions and statistical weather predic-*
1157 *tion*, Volume 1. Massachusetts Institute of Technology, Department of Meteorology
1158 Cambridge.

1159 MacQueen, J. et al. 1967. Some methods for classification and analysis of multivari-
1160 ate observations. In *Proceedings of the fifth Berkeley symposium on mathematical*

- 1161 *statistics and probability*, Volume 1, pp. 281–297. Oakland, CA, USA.
- 1162 Margules, M. 1903. Über die energie der stürme. *Jahrbuch der Zentralanstalt für*
1163 *Meteorologie, Wien* 48: Anhang .
- 1164 Marrafon, V.H., M.S. Reboita, R.P. da Rocha, and E.M. de Jesus. 2022. Classificação
1165 dos tipos de ciclones sobre o oceano atlântico sul em projeções com o regcm4 e
1166 mcgs. *Revista Brasileira de Climatologia* 30: 1–25 .
- 1167 McErlich, C., A. McDonald, J. Renwick, and A. Schuddeboom. 2023. An assessment
1168 of extra-tropical cyclone precipitation extremes over the southern hemisphere using
1169 era5. *Geophysical Research Letters* 50(17): e2023GL104130 .
- 1170 Michaelides, S. 2021. Lorenz atmospheric energy cycle in climatic projections.
1171 *Climate* 9(12): 180 .
- 1172 Michaelides, S.C. 1987. Limited area energetics of genoa cyclogenesis. *Monthly*
1173 *Weather Review* 115(1): 13–26 .
- 1174 Michaelides, S.C. 1992. A spatial and temporal energetics analysis of a baroclinic
1175 disturbance in the mediterranean. *Monthly weather review* 120(7): 1224–1243 .
- 1176 Michaelides, S.C., N.G. Prezerakos, and H.A. Flocas. 1999. Quasi-lagrangian ener-
1177 getics of an intense mediterranean cyclone. *Quarterly Journal of the Royal*
1178 *Meteorological Society* 125(553): 139–168 .
- 1179 Muench, H.S. 1965. On the dynamics of the wintertime stratosphere circulation.
1180 *Journal of Atmospheric Sciences* 22(4): 349–360 .
- 1181 Novak, L. and R. Tailleux. 2018. On the local view of atmospheric available potential
1182 energy. *Journal of the Atmospheric Sciences* 75(6): 1891–1907 .
- 1183 Okajima, S., H. Nakamura, and Y. Kaspi. 2021. Cyclonic and anticyclonic contribu-
1184 tions to atmospheric energetics. *Scientific reports* 11(1): 13202 .
- 1185 Oort, A.H. 2018. On estimates of the atmospheric energy cycle, *Renewable Energy*,
1186 Vol1.214–Vol1.235. Routledge.
- 1187 Pan, Y., L. Li, X. Jiang, G. Li, W. Zhang, X. Wang, and A.P. Ingersoll. 2017. Earth’s
1188 changing global atmospheric energy cycle in response to climate change. *Nature*
1189 *communications* 8(1): 14367 .
- 1190 Pezza, A.B., L.A. Garde, J.A.P. Veiga, and I. Simmonds. 2014. Large scale features
1191 and energetics of the hybrid subtropical low ‘duck’over the tasman sea. *Climate*
1192 *dynamics* 42: 453–466 .
- 1193 Pezza, A.B., J.A.P. Veiga, I. Simmonds, K. Keay, and M.d.S. Mesquita. 2010. Envi-
1194 ronmental energetics of an exceptional high-latitude storm. *Atmospheric Science*

- 1195 *Letters* 11(1): 39–45 .
- 1196 Rantanen, M., J. Räisänen, V.A. Sinclair, and H. Järvinen. 2019. Sensitivity
1197 of idealised baroclinic waves to mean atmospheric temperature and meridional
1198 temperature gradient changes. *Climate Dynamics* 52: 2703–2719 .
- 1199 Read, P., D. Kennedy, N. Lewis, H. Scolan, F. Tabataba-Vakili, Y. Wang, S. Wright,
1200 and R. Young. 2020. Baroclinic and barotropic instabilities in planetary atmo-
1201 spheres: energetics, equilibration and adjustment. *Nonlinear Processes in Geo-*
1202 *physics* 27(2): 147–173 .
- 1203 Reboita, M.S., R.P. Da Rocha, T. Ambrizzi, and S. Sugahara. 2010. South atlantic
1204 ocean cyclogenesis climatology simulated by regional climate model (regcm3).
1205 *Climate Dynamics* 35: 1331–1347 .
- 1206 Reboita, M.S., R.P. da Rocha, M.R. de Souza, and M. Llopart. 2018. Extratropi-
1207 cal cyclones over the southwestern south atlantic ocean: Hadgem2-es and regcm4
1208 projections. *International Journal of Climatology* 38(6): 2866–2879 .
- 1209 Reboita, M.S., M.A. Gan, R. Da Rocha, T. Ambrizzi, et al. 2010. Precipitation regimes
1210 in south america: a bibliography review. *Revista Brasileira de Meteorologia* 25(2):
1211 185–204 .
- 1212 Rudeva, I. and S.K. Gulev. 2007. Climatology of cyclone size characteristics and their
1213 changes during the cyclone life cycle. *Monthly Weather Review* 135(7): 2568–2587 .
- 1214 Sinclair, M.R. 1994. An objective cyclone climatology for the southern hemi-
1215 sphere. *Monthly Weather Review* 122(10): 2239–2256. [https://doi.org/10.1175/
1216 1520-0493\(1994\)122<2239:AOCCT>2.0.CO;2](https://doi.org/10.1175/1520-0493(1994)122<2239:AOCCT>2.0.CO;2) .
- 1217 Sinclair, M.R. 1995. A climatology of cyclogenesis for the southern hemisphere.
1218 *Monthly Weather Review* 123(6): 1601–1619 .
- 1219 Smith, P.J. 1980. The energetics of extratropical cyclones. *Reviews of Geo-*
1220 *physics* 18(2): 378–386 .
- 1221 Tecchio, R., D.C. de Souza, M.B.L. da Silva, M.C. de Oliveria Costa, R. de Camargo,
1222 and J. Harari. 2024. Mean sea level, tidal components and surges in guanabara bay
1223 (rio de janeiro) from 1990 to 2021. *International Journal of Climatology* .
- 1224 Trigo, I.F. 2006. Climatology and interannual variability of storm-tracks in the euro-
1225 atlantic sector: a comparison between era-40 and ncep/ncar reanalyses. *Climate*
1226 *Dynamics* 26(2): 127–143 .
- 1227 Veiga, J.A.P. and T. Ambrizzi. 2013. A global and hemispherical analysis of the
1228 lorenz energetics based on the representative concentration pathways used in cmip5.
1229 *Advances in Meteorology* 2013(1): 485047 .

- 1230 Veiga, J.A.P., A.B. Pezza, I. Simmonds, and P.L. Silva Dias. 2008. An analysis of the
1231 environmental energetics associated with the transition of the first south atlantic
1232 hurricane. *Geophysical Research Letters* 35(15) .
- 1233 Vincent, D. and L. Chang. 1973. Some further considerations concerning energy
1234 budgets of moving systems. *Tellus* 25(3): 224–232 .
- 1235 Wahab, M.A., H.A. Basset, and A. Lasheen. 2002. On the mechanism of winter cyclo-
1236 genesis in relation to vertical axis tilt. *Meteorology and Atmospheric Physics* 81(1):
1237 103–127 .
- 1238 Wiin-Nielsen, A. 1968. On the intensity of the general circulation of the atmosphere.
1239 *Reviews of Geophysics* 6(4): 559–579 .
- 1240 Young, R. 2014. The lorenz energy cycle in simulated rotating annulus flows. *Physics*
1241 *of Fluids* 26(5) .
- 1242 Zheng, Z. 2021. zzheng93/pyeof: First release (v0.0.0).

1243 **Supplementary information**

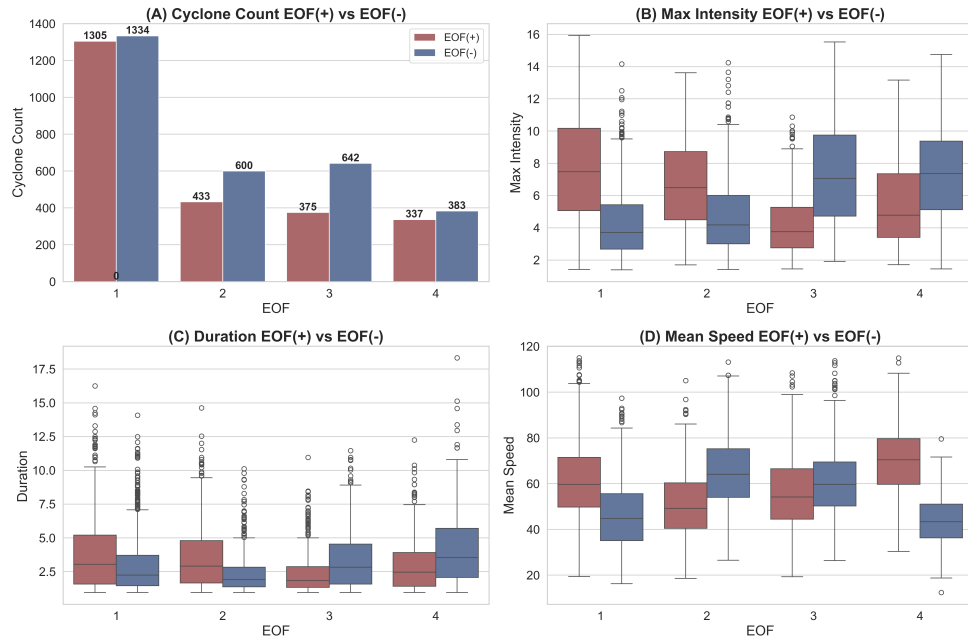


Fig. S1 Statistics for the cyclones classified as EOF(+) (red) and EOF(-) (blue). (A) Cyclone counts; (B) maximum intensity, expressed by the *minimum* central relative vorticity at 850 hPa; (C) total life-cycle duration (days); and (D) mean translational speed (km h^{-1}).

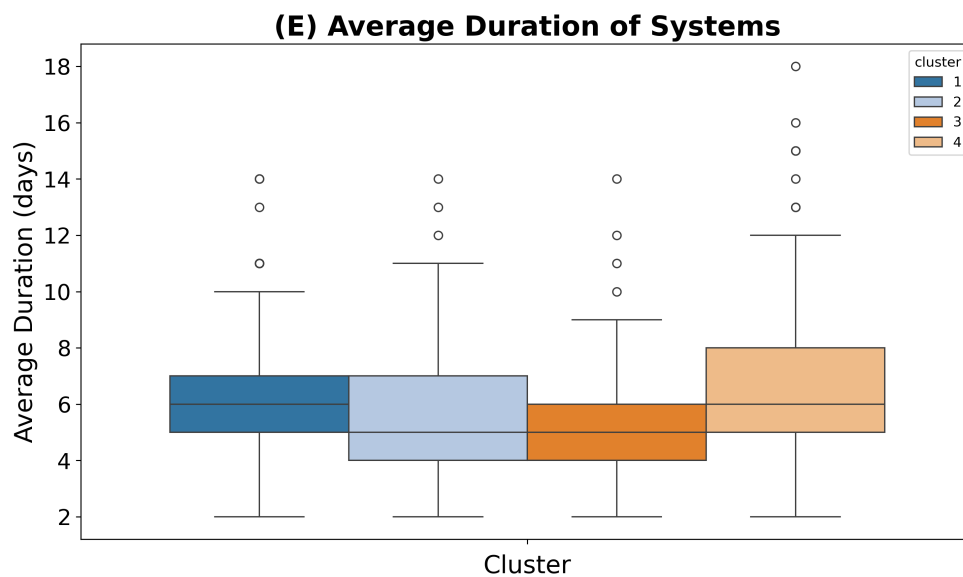


Fig. S2 Boxplot of the total life cycle-Legacy four-group duration (in days) for the four-panel retained as a placeholder. The canonical corrected analysis contains five clusters(1-4) identified among the most intense cyclones in the dataset; this graphic must be replaced by a five-group duration panel before publication.

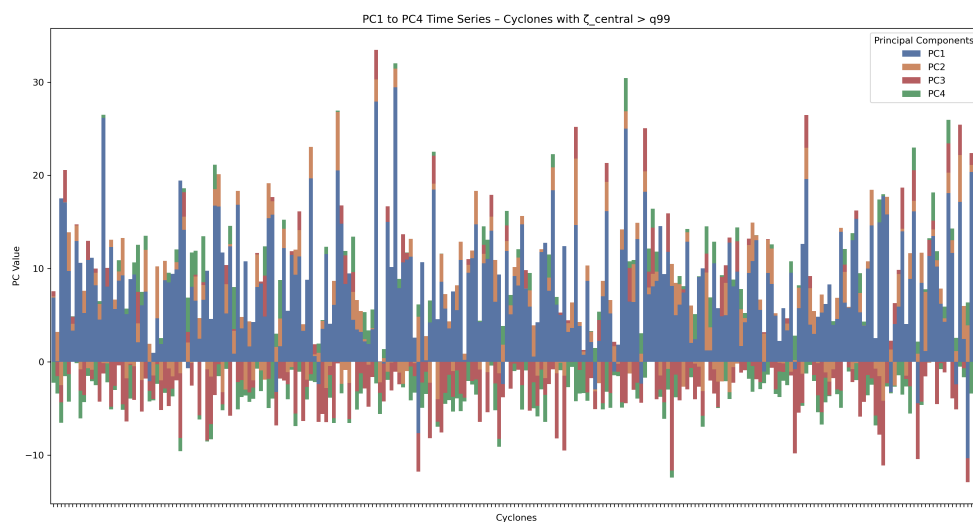


Fig. S3 Stacked time series of the first four principal components (PC1-PC4) for each intense cyclone ($\zeta_{850, \text{max}} > q99$, $\zeta_{850, \text{max}} > q90$). Positive values are plotted upward and negative values downward, so the coloured bars represent the signed contribution of each EOF mode to the energetics of every system. The x-axis lists the individual cyclones (sorted arbitrarily), while the y-axis gives the PC amplitude (dimensionless).

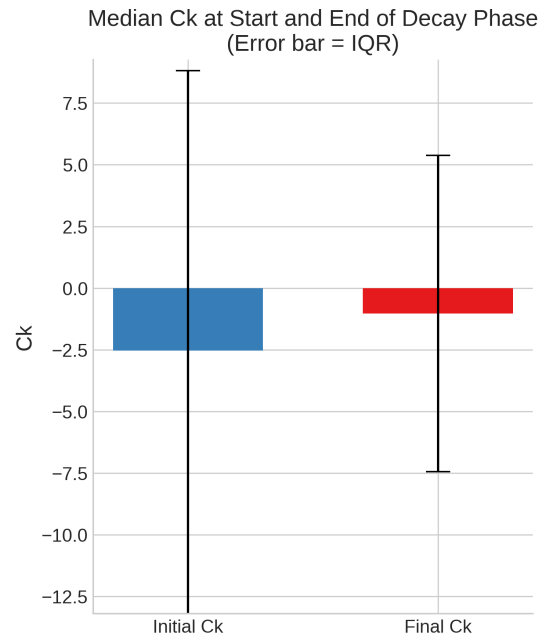


Fig. S4 Median barotropic-conversion term (C_k) at the start (blue) and end (red) of the decay phase, with the error bars representing the inter-quartile range.